

Matrices and Determinants

Question1

If $A = \begin{bmatrix} -1 & x & -3 \\ 2 & 4 & z \\ y & 5 & -6 \end{bmatrix}$ is a symmetric matrix and $B = \begin{bmatrix} 0 & 2 & q \\ p & 0 & -4 \\ -3 & r & s \end{bmatrix}$ is a skew-symmetric matrix, then $|A| + |B| - |AB| =$

AP EAPCET 2025 - 26th May Morning Shift

Options:

A.

$$xyz + pqr$$

B.

$$xyz + q + r$$

C.

$$\frac{xyz}{pq}$$

D.

$$xyz + pq + rs$$

Answer: B

Solution:

$$A = \begin{vmatrix} -1 & x & -3 \\ 2 & 4 & z \\ y & 5 & -6 \end{vmatrix} \text{ is a symmetric}$$

$$\text{So, } A = A^T$$

$$\text{that means, } x = 2, y = -3, z = 5$$

and $B = \begin{bmatrix} 0 & 2 & q \\ p & 0 & -4 \\ -3 & r & s \end{bmatrix}$ is a skew-matrix

So, $B = -B^T$

that means, $p = -2q = 3, r = 4, s = 0$

$$\begin{aligned} \text{Therefore, } |A| &= \begin{vmatrix} -1 & 2 & -3 \\ 2 & 4 & 5 \\ -3 & 5 & -6 \end{vmatrix} \\ &= -1(-49) - 2(3) - 3(22) = -23 \end{aligned}$$

$$\begin{aligned} \text{and } |B| &= \begin{vmatrix} 0 & 2 & 3 \\ -2 & 0 & -4 \\ -3 & 4 & 0 \end{vmatrix} \\ &= -2(-12) + 3(-8) = 0 \end{aligned}$$

$$\text{Now, } |AB| = |A||B| = -23 \times 0 = -23$$

therefore, $|A| + |B| - |AB| = -23$ and from the given options

$$xyz + q + r = 2 \times (-3) \times (5) + 3 + 4 = -23$$

$$\text{Hence, } |A| + |B| - |AB| = xyz + q + r$$

Question2

If the inverse of $\begin{bmatrix} -x & 14x & 7x \\ 0 & 1 & 0 \\ x & -4x & -2x \end{bmatrix}$ is $\begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$, then

$$\begin{vmatrix} x & x+1 & x+2 \\ x+1 & x+2 & x+3 \\ x+2 & x+3 & x+4 \end{vmatrix} =$$

AP EAPCET 2025 - 26th May Morning Shift

Options:

A.

$$\frac{x}{5}$$

B.

$$x - 5$$

C.

$$5x - 1$$

D.

$$x + 5$$

Answer: C

Solution:

$$A = \begin{bmatrix} -x & 14x & 7x \\ 0 & 1 & 0 \\ x & -4x & -2x \end{bmatrix}$$

$$\text{and } B = \begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$$

If $A^{-1} = B \Rightarrow$ So, $AB = I$

$$\Rightarrow \begin{bmatrix} -x & 14x & 7x \\ 0 & 1 & 0 \\ x & -4x & -2x \end{bmatrix} \begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix} = I$$
$$\Rightarrow -2x + 7x = 1 \Rightarrow x = 1/5$$

$$\text{Now, } \begin{bmatrix} x & x+1 & x+2 \\ x+1 & x+2 & x+3 \\ x+2 & x+3 & x+4 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_2$$

$$= \begin{bmatrix} x & x+1 & x+2 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

therefore determinant of the given matrix = 0

Hence, from the given options put $x = 1/5$ in $5x - 1$ we get, $5 \times \frac{1}{5} - 1 = 0$

that means,

$$\begin{vmatrix} x & x+1 & x+2 \\ x+1 & x+2 & x+3 \\ x+2 & x+3 & x+4 \end{vmatrix} = 5x - 1$$

Question3

If the system of equations

$2x + 3y - 3z = 3, x + 2y + 0z = 12x - y + z = \beta$ has infinitely many solutions, then $\frac{\alpha}{\beta} - \frac{\beta}{\alpha} =$

AP EAPCET 2025 - 26th May Morning Shift

Options:

A.

$$\frac{53}{14}$$

B.

$$\frac{45}{14}$$

C.

$$-\frac{53}{14}$$

D.

$$-\frac{45}{14}$$

Answer: B

Solution:

$$2x + 3y - 3z = 3$$

$$x + 2y + \alpha z = 1$$

$$\text{and } 2x - y + z = \beta$$

$$\text{So, } A = \begin{bmatrix} 2 & 3 & -3 \\ 1 & 2 & \alpha \\ 2 & -1 & 1 \end{bmatrix}$$

for infinite many solutions put $|A| = 0$

$$\Rightarrow 2(2 + \alpha) - 3(1 - 2\alpha) - 3(-1 - 4) = 0$$

$$\Rightarrow 1 + 8\alpha + 15 = 0 \Rightarrow \alpha = -2$$

$$\text{Now, } D_y = \begin{vmatrix} 2 & 3 & -3 \\ 1 & 1 & -2 \\ 2 & \beta & 1 \end{vmatrix}$$

$$\text{put } D_y = 0$$

$$\Rightarrow 2(1 + 2\beta) - 3(1 + 4) - 3(\beta - 2) = 0$$

$$\Rightarrow \beta - 7 = 0 \Rightarrow \beta = 7$$

$$\begin{aligned} \text{Therefore, } \frac{\alpha}{\beta} - \frac{\beta}{\alpha} &= \frac{-2}{7} - \frac{7}{(-2)} \\ &= \frac{-4 + 49}{14} = \frac{45}{14} \end{aligned}$$

Question4

If B is the inverse of a third order matrix A and $\det B = k$, then $(\text{adj}(\text{adj } A))^{-1} =$

AP EAPCET 2025 - 26th May Evening Shift

Options:

A.

kB

B.

$\frac{1}{k}B$

C.

kB^{-1}

D.

$B + kl$

Answer: A

Solution:

We have,

$$B = A^{-1}$$

$$\text{and } |B| = k \Rightarrow |A^{-1}| = k$$

$$\Rightarrow \frac{1}{|A|} = k \Rightarrow |A| = \frac{1}{k}.$$

$$\therefore (\text{adj}(\text{adj } A))^{-1} = (|A|^{n-2}A)^{-1} \text{ and } n = 3$$

$$= (|A|^1 \cdot A)^{-1} = \left(\frac{1}{k}A\right)^{-1} = kB.$$

Question5

If $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ and α, β, γ are the roots of the equation represented by $|A - xI| = 0$, then $\alpha^2 + \beta^2 + \gamma^2 =$

AP EAPCET 2025 - 26th May Evening Shift

Options:

A.

50

B.

29

C.

17

D.

27

Answer: D

Solution:

$$A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

$$\text{and } |A - XI| = 0$$

$\Rightarrow$ Characteristic equation of matrix A

$$\therefore \text{tr}(A) = 2 + 3 + 2 = 7$$

Sum of minor of diagonal elements

$$= (4) + (3) + 4 = 11$$

$$|A| = 2(4) - 2(1) + 1(-1)$$

$$= 8 - 2 - 1 = 5$$

$$\therefore |A - xI| = 0$$

$$\Rightarrow x^3 - 7x^2 + 11x - 5 = 0$$

$$\therefore \alpha + \beta + \gamma = 7$$

$$\text{and } \alpha\beta + \beta\gamma + \gamma\alpha = 11$$

$$\text{and } \alpha\beta\gamma = 5$$

$$\therefore \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$$

$$= 49 - 2 \times 11 = 49 - 22 = 27.$$

Question6

If the values of x, y and z which satisfy the equations $2x - 3y + 2z + 15 = 0$, $3x + y - z + 2 = 0$ and $x - 3y - 3z + 8 = 0$ simultaneously are α, β and γ respectively, then

AP EAPCET 2025 - 26th May Evening Shift

Options:

A.

$$\beta + \gamma = \alpha$$

B.

$$\alpha + \beta = 2\gamma$$

C.

$$2\alpha + \beta = \gamma$$

D.

$$2\beta + \gamma = 2\alpha$$

Answer: C

Solution:

$$\text{Given, } 2x - 3y + 2z = -15$$

$$3x + y - z = -2$$

$$x - 3y - 3z = -8$$

$$\begin{aligned}\therefore \Delta &= \begin{vmatrix} 2 & -3 & 2 \\ 3 & 1 & -1 \\ 1 & -3 & -3 \end{vmatrix} \\ &= 2(-3 - 3) + 3(-9 + 1) + 2(-9 - 1) \\ &= -12 - 24 - 20 = -56\end{aligned}$$

$$\begin{aligned}\Delta_x &= \begin{vmatrix} -15 & -3 & 2 \\ -2 & 1 & -1 \\ -8 & -3 & -3 \end{vmatrix} \\ &= -15(-3 - 3) + 3(6 - 8) + 2(6 + 8) \\ &= 90 - 6 + 28 = 112\end{aligned}$$

$$\begin{aligned}\Delta_y &= \begin{vmatrix} 2 & -15 & 2 \\ 3 & -2 & -1 \\ 1 & -8 & -3 \end{vmatrix} \\ &= 2(6 - 8) + 15(-9 + 1) + 2(-24 + 2) \\ &= -4 - 120 - 44 = -168\end{aligned}$$

$$\Delta_z = \begin{vmatrix} 2 & -3 & -15 \\ 3 & 1 & -2 \\ 1 & -3 & -8 \end{vmatrix}$$

$$= 2(-8 - 6) + 3(-24 + 2) - 15(-9 - 1) \\ = -28 - 66 + 150 = 150 - 94 = 56.$$

$$\alpha = \frac{\Delta_x}{\Delta}, \beta = \frac{\Delta_y}{\Delta} \text{ and } \gamma = \frac{\Delta_z}{\Delta}$$

$$\Rightarrow \alpha = \frac{112}{-56}, \beta = \frac{-168}{-56} \text{ and } \gamma = \frac{56}{-56}$$

$$\Rightarrow \alpha = -2, \beta = 3 \text{ and } \gamma = -1$$

$$\therefore 2\alpha + \beta = \gamma$$

Question7

If a is the determinant of the adjoint of the matrix $\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 3 \end{bmatrix}$ and b is the determinant of the inverse of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & -3 & -1 \\ 2 & 1 & -4 \end{bmatrix}$, then $\frac{b+1}{18b} =$

AP EAPCET 2025 - 24th May Morning Shift

Options:

A.

$$a$$

B.

$$10a$$

C.

$$2 + a$$

D.

$$2a$$

Answer: A

Solution:

$$\text{Let } A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 3 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & -3 & -1 \\ 2 & 1 & -4 \end{bmatrix}$$

Also, $a = \det(\text{adj}(A))$ and $b = \det(B^{-1})$

Now, $a = \det(\text{adj}(A)) = (\det(A))^{n-1}$ for an $n \times n$ matrix.

Since A is 3×3 matrix, so $a = (\det(A))^{3-1} = (\det(A))^2$

$$\begin{aligned}\det A &= 1 \cdot \begin{vmatrix} 2 & 3 \\ 3 & 3 \end{vmatrix} - 1 \cdot \begin{vmatrix} 1 & 3 \\ 2 & 3 \end{vmatrix} + 2 \cdot \begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix} \\ &= 1 \cdot (6 - 9) - 1(3 - 6) + 2(3 - 4) \\ &= -3 + 3 - 2 = -2 \\ a &= (\det(A))^2 = (-2)^2 = 4 \quad \dots (i)\end{aligned}$$

$$\begin{aligned}\text{Now, } \det(B) &= \begin{vmatrix} 1 & 2 & 3 \\ 4 & -3 & -1 \\ 2 & 1 & -4 \end{vmatrix} \\ &= 1 \cdot \begin{vmatrix} -3 & -1 \\ 1 & -4 \end{vmatrix} - 2 \begin{vmatrix} 4 & -1 \\ 2 & -4 \end{vmatrix} + 3 \cdot \begin{vmatrix} 4 & -3 \\ 2 & 1 \end{vmatrix} \\ &= (12 + 1) - 2(-16 + 2) + 3(4 + 6) \\ &= 13 + 28 + 30 = 71\end{aligned}$$

$$\text{So, } b = \det(B^{-1}) = \frac{1}{\det(B)} = \frac{1}{71}$$

$$\begin{aligned}b &= \frac{1}{71} \\ \Rightarrow b + 1 &= \frac{1 + 71}{71} = \frac{72}{71} \\ \text{Then, } \frac{b + 1}{18b} &= \frac{72/71}{18 \times \frac{1}{71}} \\ &= \frac{72}{71} \times \frac{71}{18} = \frac{72}{18} = 4 = a \quad [\text{using Eq. (i)}] \\ \therefore \frac{b + 1}{18b} &= a\end{aligned}$$

Question8

Consider two systems of 3 linear equations in 3 unknowns $AX = B$ and $CX = D$. If $AX = B$ has unique solution D and $CX = D$ has unique solution B , then the solution of $(A - C^{-1})X = 0$ is

AP EAPCET 2025 - 24th May Morning Shift

Options:

A.

B

B.

D

C.

$B + D$

D.

$B - D$

Answer: B

Solution:

Given, $AX = B$ and $CX = D$ and $AX = B$ has unique solution D and $CX = D$ has unique solution B .

Since, $X = D$ is the unique solution of $AX = B$

$$\therefore AD = B \quad \dots (i)$$

And, $X = B$ is the unique solution of $CX = D$,

$$\therefore CB = D \Rightarrow B = C^{-1}D \quad \dots (ii)$$

From Eq. (i) and (ii),

$$\begin{aligned} AD &= C^{-1}D \\ (A - C^{-1})D &= 0 \end{aligned}$$

So, the solution of $(A - C^{-1})X = 0$ is D

Question9

$f(x)$ is an n th degree polynomial satisfying

$$f(x) = \frac{1}{2} \begin{vmatrix} f(x) & f\left(\frac{1}{x}\right) - f(x) \\ 1 & f\left(\frac{1}{x}\right) \end{vmatrix}. \text{ If } f(2) = 33, \text{ then the value of } f(3) \text{ is}$$

AP EAPCET 2025 - 24th May Morning Shift

Options:

A.

126

B.

214

C.

244

D.

-124

Answer: C

Solution:

Given,

$$f(x) = \frac{1}{2} \begin{vmatrix} f(x) & f\left(\frac{1}{x}\right) - f(x) \\ 1 & f\left(\frac{1}{x}\right) \end{vmatrix}$$

and $f(2) = 33$

$$\text{Here, } f(x) = \frac{1}{2} [f(x) \cdot f\left(\frac{1}{x}\right) - f\left(\frac{1}{x}\right) + f(x)] \quad \dots (i)$$

Let

$$f(x) = a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n \\ (a_0 \neq 0)$$

Putting this value of $f(x)$ in Eq. (i),

$$f(x) = \frac{1}{2} \left[f(x) \cdot f\left(\frac{1}{x}\right) - f\left(\frac{1}{x}\right) + f(x) \right] \\ \Rightarrow f(x) \cdot f\left(\frac{1}{x}\right) = f\left(\frac{1}{x}\right) + f(x)$$

$$\text{Hence, } f(x) = x^n + 1 \text{ or } -x^n + 1$$

$$\text{Now, } f(2) = 2^n + 1 = 33$$

$$\Rightarrow 2^n = 32$$

$$\Rightarrow 2^n = 2^5$$

$$\Rightarrow n = 5$$

$$\text{And } f(2) = -2^n + 1 = 33$$

$$\Rightarrow -2^n = 33 - 1 = 32 = 2^5$$

which is not possible.

$$\therefore f(x) = x^n + 1 = x^5 + 1$$

$$\text{So, } f(3) = 3^5 + 1 = 243 + 1 = 244$$

Question10

If $P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}$ is the adjoint of a matrix A and $\det A = 4$, then the value of α is

AP EAPCET 2025 - 23rd May Evening Shift

Options:

A.

3

B.

22

C.

11

D.

4

Answer: C

Solution:

$$\because P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}$$

$$\begin{aligned} \because \operatorname{adj}(A) &= (\det A) \cdot A^{-1} \\ \Rightarrow \det(\operatorname{adj}(A)) &= (\det A)^{n-1} \end{aligned}$$

So, for $n = 3$

$$\begin{aligned} \det(\operatorname{adj}(A)) &= (\det A)^{3-1} \\ &= 4^2 = 16 \quad \dots (i) \end{aligned}$$

$$\begin{aligned} \text{and } \det(A) &= 1(12 - 12) - \alpha(4 - 6) + 3(4 - 6) \\ &= 2\alpha - 6 \quad \dots (ii) \end{aligned}$$

Hence, $2\alpha - 6 = 16$

$$\Rightarrow 2\alpha = 22$$

$$\Rightarrow \alpha = 11$$

Question11

If α is a real root of the equation $x^3 + 6x^2 + 5x - 42 = 0$, then the determinant of the matrix

$$\begin{bmatrix} \alpha - 1 & \alpha + 1 & \alpha + 2 \\ \alpha - 2 & \alpha + 3 & \alpha - 3 \\ \alpha + 4 & \alpha - 4 & \alpha + 5 \end{bmatrix} \text{ is}$$

AP EAPCET 2025 - 23rd May Evening Shift

Options:

A.

90

B.

120

C.

-105

D.

-135

Answer: C

Solution:

We are given the equation $x^3 + 6x^2 + 5x - 42 = 0$.

Let's check if $x = 2$ is a root by plugging it in:

$$(2)^3 + 6 \times (2)^2 + 5 \times 2 - 42 = 8 + 24 + 10 - 42 = 42 - 42 = 0$$

Since the equation equals zero, $x = 2$ is a real root. So, $\alpha = 2$.

Now, we substitute $\alpha = 2$ into the matrix:

$$\begin{bmatrix} \alpha - 1 & \alpha + 1 & \alpha + 2 \\ \alpha - 2 & \alpha + 3 & \alpha - 3 \\ \alpha + 4 & \alpha - 4 & \alpha + 5 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 4 \\ 0 & 5 & -1 \\ 6 & -2 & 7 \end{bmatrix}$$

Next, let's find the determinant of this matrix:

$$\begin{aligned}
&= \begin{vmatrix} 1 & 3 & 4 \\ 0 & 5 & -1 \\ 6 & -2 & 7 \end{vmatrix} \\
&= 1 \times (5 \times 7 - (-1) \times (-2)) - 3 \times (0 \times 7 - (-1) \times 6) + 4 \times (0 \times (-2) - 5 \times 6) \\
&= 1 \times (35 - 2) - 3 \times (0 + 6) + 4 \times (0 - 30) \\
&= 33 - 18 - 120 \\
&= -105
\end{aligned}$$

Question12

The rank of the matrix $\begin{bmatrix} 2 & -3 & 4 & 0 \\ 5 & -4 & 2 & 1 \\ 1 & -3 & 5 & -4 \end{bmatrix}$ is

AP EAPCET 2025 - 23rd May Evening Shift

Options:

A.

0

B.

3

C.

2

D.

1

Answer: B

Solution:

Given matrix

$$\begin{bmatrix} 2 & -3 & 4 & 0 \\ 5 & -4 & 2 & 1 \\ 1 & -3 & 5 & -4 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - \frac{5}{2}R_1 \text{ and } R_3 \rightarrow R_3 - \frac{1}{2}R_1$$

$$= \begin{bmatrix} 2 & -3 & 4 & 0 \\ 0 & 3.5 & -8 & 1 \\ 0 & -1.5 & 3 & -4 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + \frac{1.5}{3.5}R_2 = R_3 + \frac{3}{7}R_2$$

$$= \begin{bmatrix} 2 & -3 & 4 & 0 \\ 0 & 3.5 & -8 & 1 \\ 0 & 0 & -3/7 & 25/7 \end{bmatrix}$$

$\therefore$ All 3 rows are non-zero

Hence, rank of the matrix = 3

Question13

$$A = \begin{bmatrix} 0 & k & k \\ k & -4 & -6 \\ k & -3 & -5 \end{bmatrix} \text{ is a singular matrix for}$$

AP EAPCET 2025 - 23rd May Morning Shift

Options:

A.

$k = 2$ only

B.

$k = \pm 2$ only

C.

no real value of k

D.

all real values of k

Answer: D

Solution:

Given, $A = \begin{bmatrix} 0 & k & k \\ k & -4 & -6 \\ k & -3 & -5 \end{bmatrix}$

$\therefore A$ is an singular matrix.

$\therefore |A| = 0$

$\Rightarrow \begin{vmatrix} 0 & k & k \\ k & -4 & -6 \\ k & -3 & -5 \end{vmatrix} = 0$

$\Rightarrow 0 - k(-5k + 6k) + k(-3k + 4k) = 0$

$\Rightarrow -k^2 + k^2 = 0$ which is true for all real values of k .

$\therefore k \in R$

Question14

If $A = \begin{bmatrix} 1 & 2 & x \\ 4 & -1 & 7 \\ 2 & 4 & -6 \end{bmatrix}$ and the rank of A is 2 , then the value of x is equal to

AP EAPCET 2025 - 23rd May Morning Shift

Options:

A.

1

B.

0

C.

-3

D.

3

Answer: C

Solution:

Given, matrix $A = \begin{bmatrix} 1 & 2 & x \\ 4 & -1 & 7 \\ 2 & 4 & -6 \end{bmatrix}$ and the rank of $A = 2$

$$= 1(6 - 28) - 2(-24 - 14) + x(16 + 2)$$

$$|A| = 54 + 18x$$

Since, the rank of A is 2 then the determinant of A must be zero.

So, we set the determinant equal to zero and solve for x .

$$\text{i.e. } 54 + 18x = 0 \Rightarrow x = -3$$

Question15

$$\begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix} + \begin{vmatrix} 1 & \frac{1}{3} \\ 3 & 1 \end{vmatrix} + \begin{vmatrix} \frac{1}{2} & \frac{1}{9} \\ 3 & 1 \end{vmatrix} + \begin{vmatrix} \frac{1}{4} & \frac{1}{27} \\ 3 & 1 \end{vmatrix} + \dots \infty =$$

AP EAPCET 2025 - 23rd May Morning Shift

Options:

A.

0

B.

1/2

C.

-1/2

D.

-1

Answer: C

Solution:

$$\begin{aligned} & \begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix} + \begin{vmatrix} 1 & \frac{1}{3} \\ 3 & 1 \end{vmatrix} + \begin{vmatrix} \frac{1}{2} & \frac{1}{9} \\ 3 & 1 \end{vmatrix} + \begin{vmatrix} \frac{1}{4} & \frac{1}{27} \\ 3 & 1 \end{vmatrix} + \dots + \infty \\ &= (2 - 3) + \left(1 - 3 \times \frac{1}{3}\right) + \left(\frac{1}{2} - 3 \times \frac{1}{9}\right) + \left(\frac{1}{4} - 3 \times \frac{1}{27}\right) + \left(\frac{1}{8} - 3 \times \frac{1}{81}\right) + \dots + \infty \\ &= -1 + 0 + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{2^2} - \frac{1}{3^2}\right) + \left(\frac{1}{2^3} - \frac{1}{3^3}\right) + \dots + \infty \end{aligned}$$

$$\begin{aligned}
&= -1 + \left[\left(\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \dots + \infty \right) - \left(\frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \frac{1}{3^4} + \dots + \infty \right) \right] \\
&= -1 + \left(\frac{\frac{1}{2}}{1 - \frac{1}{2}} \right) - \left(\frac{\frac{1}{3}}{1 - \frac{1}{3}} \right) \\
&\text{(Using sum of infinite GP formula)} \\
&= -1 + \frac{\frac{1}{2}}{\frac{1}{2}} - \frac{\frac{1}{3}}{\frac{2}{3}} = -1 + 1 - \frac{1}{2} = -\frac{1}{2}
\end{aligned}$$

Question16

If $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 2 & 1 & 6 \end{bmatrix}$ and $|\text{adj}(\text{adj } A)|(\text{adj } A)^{-1} = kA$, then $k =$

AP EAPCET 2025 - 22nd May Evening Shift

Options:

A.

1296

B.

216

C.

36

D.

432

Answer: B

Solution:

$$|A| = 1(18 - 5) - 2(6 - 10) + 3(1 - 6)$$

$$= 13 + 8 - 15 = 6$$

$$|\text{adj}(\text{adj } A)| = |A|^4 = 6^4$$

$$\Rightarrow (\text{adj } A)^{-1} = \frac{1}{6} A$$

$$\Rightarrow |\text{adj}(\text{adj } A)|(\text{adj } A)^{-1} = kA$$

$$\Rightarrow 6^4 \cdot \frac{1}{6} A = kA$$

$$\therefore k = 216$$

Question17

If the values $x = \alpha, y = \beta, z = \gamma$ satisfy all the 3 equations

$x + 2y + 3z = 4, 3x + y + z = 3$ and $x + 3y + 3z = 2$, then $3\alpha + \gamma =$

AP EAPCET 2025 - 22nd May Evening Shift

Options:

A.

$$\beta$$

B.

$$2\beta$$

C.

$$1 - 2\beta$$

D.

$$2\beta + 1$$

Answer: C

Solution:

$$\alpha + 2\beta + 3\gamma = 4 \quad \dots (i)$$

$$3\alpha + \beta + \gamma = 3 \quad \dots (ii)$$

$$\alpha + 3\beta + 3\gamma = 2 \quad \dots (iii)$$

$$\text{From Eqs. (ii), } 3\alpha + \gamma = 3 - \beta \quad \dots (iv)$$

Subtracting Eq. (iii) from Eq. (i)

$$\begin{array}{r}
 \alpha + 2\beta + 3\gamma = 4 \\
 \alpha + 3\beta + 3\gamma = 2 \\
 \hline
 -\beta = 2 \Rightarrow \beta = -2
 \end{array}$$

Bu Eq. (iv)

$$\begin{aligned}
 3\alpha + \gamma &= 3 - (-2) = 5 \\
 \Rightarrow 3\alpha + \gamma &= 1 - 2\beta
 \end{aligned}$$

Question18

The number of solutions of the system of equations $2x + y - z = 7, x - 3y + 2z = 1, x + 4y - 3z = 5$ is

AP EAPCET 2025 - 22nd May Evening Shift

Options:

A.

1

B.

0

C.

Infinite

D.

2

Answer: B

Solution:

$$2x + y - z = 7, x - 3y + 2z = 1,$$

$$x + 4y - 3z = 5$$

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & -3 & 2 \\ 1 & 4 & -3 \end{bmatrix}$$

$$\Rightarrow |A| = \begin{vmatrix} 2 & 1 & -1 \\ 1 & -3 & 2 \\ 1 & 4 & -3 \end{vmatrix}$$

$$= 2(9 - 8) - 1(-3 - 2) - 1(4 + 3)$$

$$= 2 + 5 - 7 = 0$$

$$\Rightarrow \text{Number of solution} = 0$$

Question19

The value of p and q is that system of equations

$2x + py + 6z = 8$, $x + 2y + qz = 5$ and $x + y + 3z = 4$ may have no solution are

AP EAPCET 2025 - 22nd May Morning Shift

Options:

A.

$$p \neq 2, q = 3$$

B.

$$p \neq 2, q \neq 3$$

C.

$$p = 2, q = \frac{15}{4}$$

D.

$$p = 2, q = 3$$

Answer: A

Solution:

$2x + py + 6z = 8$, $x + 2y + qz = 5$ and $x + y + 3z = 4$ may have no solution.

$$x = \frac{\Delta x}{\Delta}, y = \frac{\Delta y}{\Delta}, z = \frac{\Delta z}{\Delta}$$

$$\Delta = \begin{vmatrix} 2 & p & 6 \\ 1 & 2 & q \\ 1 & 1 & 3 \end{vmatrix} = (p-2)(q-3)$$

$$\Delta_x = \begin{vmatrix} 8 & p & 6 \\ 5 & 2 & q \\ 4 & 1 & 3 \end{vmatrix} = (15-4q)(2-p)$$

$$\Delta_y = \begin{vmatrix} 2 & 8 & 6 \\ 1 & 5 & q \\ 1 & 4 & 3 \end{vmatrix} = 0$$

$$\Delta_z = \begin{vmatrix} 2 & p & 8 \\ 1 & 2 & 5 \\ 1 & 1 & 4 \end{vmatrix} = p-2$$

For no solution

$$(p-2)(q-3) = 0$$

$$p-2 \neq 0$$

$$q = 3$$

$$p \neq 2$$

Question20

***A* is the set of all matrices of order 3 with entries 0 or 1 only. *B* is the subset of *A* consisting of all matrices with determinant value 1 . If *C* is the subset of *A* consisting of all matrices with determinant value -1 , then**

AP EAPCET 2025 - 22nd May Morning Shift

Options:

A.

$$A = B \cup C$$

B.

C is empty

C.

B and *C* contain the same number of elements

D.

B has twice as many elements as *C*

Answer: C

Solution:

A is det of order 3 with entries 0 or 1 .

$B \subseteq A$ and $|B| = 1$ and $C \subset A$ of and $|C| = -1$

$\therefore B$ and C has same number of elements.

Question21

Consider the matrices $A = \begin{bmatrix} x & y & 0 \\ -3 & 1 & 2 \\ 1 & -2 & z \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 & -2 \\ 2 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}$

If the cofactors of the elements $z, 1$ in 3rd row and x of A are 9, 4, 3, respectively then $AB =$

AP EAPCET 2025 - 22nd May Morning Shift

Options:

A.

$$\begin{bmatrix} -7 & -4 & -8 \\ -1 & 8 & 7 \\ 3 & -3 & -4 \end{bmatrix}$$

B.

$$\begin{bmatrix} 7 & -6 & -8 \\ -5 & 4 & -5 \\ -5 & -3 & -4 \end{bmatrix}$$

C.

$$\begin{bmatrix} 7 & -6 & -4 \\ 3 & 8 & 7 \\ -5 & -3 & -4 \end{bmatrix}$$

D.

$$\begin{bmatrix} 7 & -6 & 8 \\ -1 & 8 & -5 \\ 3 & -3 & -4 \end{bmatrix}$$

Answer: C

Solution:

$$(c) A = \begin{bmatrix} x & y & 0 \\ -3 & 1 & 2 \\ 1 & -2 & z \end{bmatrix} \text{ and}$$

$$B = \begin{bmatrix} 1 & -2 & -2 \\ 2 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}$$

Cofactor of $z = 9$, cofactor of 1 in 3rd row of $A = 4$

$$\Rightarrow \begin{matrix} x + 3y = 9 \\ x = 3 \end{matrix} \quad \dots (i)$$

$$\Rightarrow \begin{matrix} 2y = 4 \\ y = 2 \end{matrix} \quad \dots (ii)$$

Cofactor of $x = 3$

$$z + 4 = 3 \Rightarrow z = -1 \quad \dots (iii)$$

$$\begin{aligned} \therefore AB &= \begin{bmatrix} 3 & 2 & 0 \\ -3 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & -2 & -2 \\ 2 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 7 & -6 & -4 \\ 3 & 8 & 7 \\ -5 & -3 & -4 \end{bmatrix} \end{aligned}$$

Question22

$$\text{If } A = \begin{bmatrix} 1 & 2 & -2 \\ 2 & -1 & 2 \\ -1 & 1 & -2 \end{bmatrix}, \text{ then } A + 2A^{-1} =$$

AP EAPCET 2025 - 21st May Evening Shift

Options:

A.

$$\begin{bmatrix} 1 & 4 & 0 \\ 4 & -5 & -4 \\ 0 & -2 & -7 \end{bmatrix}$$

B.

$$\begin{bmatrix} 0 & 2 & 2 \\ 2 & -4 & -6 \\ 2 & -3 & -5 \end{bmatrix}$$

C.

$$\begin{bmatrix} 0 & 2 & 1 \\ 2 & -4 & -3 \\ 2 & -6 & -5 \end{bmatrix}$$

D.

$$\begin{bmatrix} 1 & 4 & -1 \\ 4 & -5 & -1 \\ 1 & -5 & -7 \end{bmatrix}$$

Answer: A

Solution:

$$\begin{aligned} \text{We have, } A &= \begin{bmatrix} 1 & 2 & -2 \\ 2 & -1 & 2 \\ -1 & 1 & -2 \end{bmatrix} \\ |A| &= |(2-2) - 2(-4+2) - 2(2-1)| = 4 - 2 = 2 \\ \text{adj}(A) &= \begin{bmatrix} 0 & 2 & 2 \\ 2 & -4 & -6 \\ 1 & -3 & -5 \end{bmatrix} \\ A^{-1} &= \frac{\text{adj}(A)}{|A|} = \frac{1}{2} \begin{bmatrix} 0 & 2 & 2 \\ 2 & -4 & -6 \\ 1 & -3 & -5 \end{bmatrix} \\ \text{Then, } A + 2A^{-1} &= \begin{bmatrix} 1 & 4 & 0 \\ 4 & -5 & -4 \\ 0 & -2 & -7 \end{bmatrix} \end{aligned}$$

Question23

$$\begin{aligned} \text{If } A &= \begin{bmatrix} a & b & c \\ d & e & f \\ l & m & n \end{bmatrix} \text{ is a matrix such that } |A| > 0 \text{ and} \\ \text{adj}(A) &= \begin{bmatrix} 0 & 4 & -6 \\ 10 & 8 & 0 \\ 2 & 4 & -4 \end{bmatrix}, \text{ then } \frac{cd}{fb} + \frac{ln}{em} = \end{aligned}$$

Options:

A.

$2a$

B.

$a + m$

C.

$a + b$

D.

a

Answer: B**Solution:**

$$\text{We have, } A = \begin{bmatrix} a & b & c \\ d & e & f \\ l & m & n \end{bmatrix} \text{ and } \text{adj}(A) = \begin{bmatrix} 0 & 4 & -6 \\ 10 & 8 & 0 \\ 2 & 4 & -4 \end{bmatrix} \mid \text{adj } A = |A|^2 \Rightarrow |A|^2 = 16 \Rightarrow |A| = 4$$

$$\text{We know that } \text{adj}(\text{adj } A) = |A|^{n-2}A = |A|^{3-2}A = |A|A$$

$$\Rightarrow A = \frac{1}{|A|} \text{adj}(\text{adj } A)$$

$$\text{Here, } \text{adj}(\text{adj } A) = \begin{bmatrix} -32 & -8 & 48 \\ 40 & 12 & -60 \\ 24 & 8 & -40 \end{bmatrix}$$

$$\therefore A = \frac{1}{4} \begin{bmatrix} -32 & -8 & 48 \\ 40 & 12 & -60 \\ 24 & 8 & -40 \end{bmatrix} = \begin{bmatrix} -8 & -2 & 12 \\ 10 & 3 & -15 \\ 6 & 2 & -10 \end{bmatrix}$$

$$\Rightarrow a = -8, b = -2, c = 12, d = 10, e = 3, f = -15, l = 6, m = 2,$$

$$n = -10$$

$$\begin{aligned} \therefore \frac{cd}{fb} + \frac{ln}{em} &= \frac{12 \times 10}{(-15) \times (-2)} + \frac{6 \times (-10)}{3 \times 2} \\ &= \frac{120}{30} - \frac{60}{6} = 4 - 10 = -6 \end{aligned}$$

$$\text{Also, (a) } 2a = -16$$

$$\text{(b) } a + m = -8 + 2 = -6$$

$$\text{(c) } a + b = -8 - 2 = -10$$

$$\text{(d) } a = -8$$

Question24

In solving a system of linear equations $AX = B$ by Cramer's rule, in the usual notation, if $\Delta_1 = \begin{vmatrix} -11 & 1 & -7 \\ -4 & 1 & -2 \\ 5 & 1 & 1 \end{vmatrix}$ and $\Delta_3 = \begin{vmatrix} 4 & 1 & -11 \\ 1 & 1 & -4 \\ 4 & 1 & 5 \end{vmatrix}$, then $X =$

AP EAPCET 2025 - 21st May Evening Shift

Options:

A.

$$\begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$$

B.

$$\begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$$

C.

$$\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

D.

$$\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

Answer: C

Solution:

$$\text{Given, } \Delta_1 = \begin{vmatrix} -11 & 1 & -7 \\ -4 & 1 & -2 \\ 5 & 1 & 1 \end{vmatrix}$$

$$\therefore \Delta \Delta_1 = -11(3) - 1(6) - 7(-9) = -33 - 6 + 63 = 24$$

$$\Delta_3 = \begin{vmatrix} 4 & 1 & -11 \\ 1 & 1 & -4 \\ 4 & 1 & 5 \end{vmatrix} = 4(9) - 1(21) - 11(-3) = 48$$

$$\Delta = \begin{vmatrix} 4 & 1 & -7 \\ 1 & 1 & -2 \\ 4 & 1 & 1 \end{vmatrix} = 4(3) - 1(9) - 7(-3) = 24$$

By inspection

$$\Delta_2 = \begin{vmatrix} 4 & -11 & -7 \\ 1 & -4 & -2 \\ 4 & 5 & 1 \end{vmatrix} = 4(6) + 11(9) - 7(21) = -24$$

$$x = \frac{\Delta_1}{\Delta} = \frac{24}{24} = 1, y = \frac{\Delta_2}{\Delta} = \frac{-24}{24} = -1 \text{ and } z = \frac{\Delta_3}{\Delta} = \frac{48}{24} = 2$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

Question 25

If A and B are both 3×3 matrices, then which of the following statements are true?

(i) $AB = 0 \Rightarrow A = 0 \text{ or } B = 0$

(ii) $AB = I_3 \Rightarrow A^{-1} = B$

(iii) $(A - B)^2 = A^2 - 2AB + B^2$

AP EAPCET 2025 - 21st May Morning Shift

Options:

A.

(i) is false and (ii), (iii) are true

B.

(ii) is true (i), (iii) are false

C.

(i) and (ii) are true, (iii) is false

D.

All are true

Answer: B

Solution:

(i) $AB = 0 \Rightarrow A = 0$ or $B = 0$

Which is false.

(ii) If $AB = BA = I_3 \Rightarrow A^{-1} = B$

$\therefore$ It is true.

$$(iii) (A - B)^2 = (A - B)(A - B) \\ = A^2 - AB - BA + B^2$$

$\Rightarrow A^2 - 2AB + B^2$ is true only when

$$AB = BA$$

$\therefore$ It is false.

Question 26

$A = \begin{bmatrix} 1 & -1 & 2 \\ -2 & 3 & -3 \end{bmatrix}$ is the given matrix and A^T represents the transpose of A , then $AA^T - A - A^T =$

AP EAPCET 2025 - 21st May Morning Shift

Options:

A.

$$\begin{bmatrix} 4 & 8 & 12 \\ 8 & 16 & -28 \\ 12 & -28 & 47 \end{bmatrix}$$

B.

$$\begin{bmatrix} 4 & -8 & 12 \\ -8 & 16 & -28 \\ 12 & -28 & 47 \end{bmatrix}$$

C.

$$\begin{bmatrix} 4 & -8 & 12 \\ -8 & 16 & 28 \\ 12 & 28 & 47 \end{bmatrix}$$

D.

$$\begin{bmatrix} 4 & -8 & -12 \\ -8 & 16 & -28 \\ -12 & -28 & 47 \end{bmatrix}$$

Answer: B

Solution:

$$\text{Given, } A = \begin{bmatrix} 1 & -1 & 2 \\ -2 & 3 & -3 \\ 4 & -4 & 5 \end{bmatrix} \text{ then}$$

$$A^T = \begin{bmatrix} 1 & -2 & 4 \\ -1 & 3 & -4 \\ 2 & -3 & 5 \end{bmatrix}$$

$$\text{Now, } AA^T = \begin{bmatrix} 6 & -11 & 18 \\ -11 & 22 & -35 \\ 18 & -35 & 57 \end{bmatrix}$$

$$\text{Then, } AA^T - A - A^T = \begin{bmatrix} 4 & -8 & 12 \\ -8 & 16 & -28 \\ 12 & -28 & 47 \end{bmatrix}$$

Question27

$$\text{If } A = \begin{bmatrix} x & 2 & 1 \\ -2 & y & 0 \\ 2 & 0 & -1 \end{bmatrix}, x \text{ and } y \text{ are non-zero numbers, trace of } A = 0$$

and determinant of $A = -6$, then the minor of the elements 1 of A is

AP EAPCET 2025 - 21st May Morning Shift

Options:

A.

B.

4

C.

2

D.

-2

Answer: A

Solution:

$$\text{We have, } A = \begin{bmatrix} x & 2 & 1 \\ -2 & y & 0 \\ 2 & 0 & -1 \end{bmatrix}$$

$$\text{trace of } (A) = x + y - 1 = 0 \quad \dots (i)$$

$$\det A = -2y - xy - 4 = -6$$

$$2y + xy = 2 \quad \dots (ii)$$

Solving Eqs. (i) and (ii), we get

$$2y + y(1 - y) = 2$$

$$\Rightarrow y^2 - 3y + 2 = 0$$

$$y = 1 \text{ and } 2$$

$$\text{and } x = 0 \text{ and } -1$$

Then, the minor of the element 1 is

$$\begin{vmatrix} -2 & y \\ 2 & 0 \end{vmatrix} \\ = -2y = -2(2) = -4$$

Question28

$$4. \text{ If } A = \begin{bmatrix} 83 & 74 & 41 \\ 93 & 96 & 31 \\ 24 & 15 & 79 \end{bmatrix}, \text{ then } \det (A - A^T) \text{ is equal to}$$

AP EAPCET 2024 - 23th May Morning Shift

Options:

A. 0

B. -7851

C. 2442

D. 1

Answer: A

Solution:

$$\begin{aligned}\text{We have, } A &= \begin{bmatrix} 83 & 74 & 41 \\ 93 & 96 & 31 \\ 24 & 15 & 79 \end{bmatrix} \\ A^T &= \begin{bmatrix} 83 & 93 & 24 \\ 74 & 96 & 15 \\ 41 & 31 & 79 \end{bmatrix} \\ \therefore A - A^T &= \begin{bmatrix} 83 & 74 & 41 \\ 93 & 96 & 31 \\ 24 & 15 & 79 \end{bmatrix} - \begin{bmatrix} 83 & 93 & 24 \\ 74 & 96 & 15 \\ 41 & 31 & 79 \end{bmatrix} \\ &= \begin{bmatrix} 0 & -19 & 17 \\ 19 & 0 & 16 \\ -17 & -16 & 0 \end{bmatrix} \\ \therefore \det(A - A^T) &= \begin{vmatrix} 0 & -19 & 17 \\ 19 & 0 & 16 \\ -17 & -16 & 0 \end{vmatrix} \\ &= 0(0 + 16 \times 16) + 19(9 \times 0 + 16 \times 17) \\ &\quad + 17(-19 \times 16 - 0) \\ &= 19 \times 16 \times 17 - 17 \times 19 \times 16 \\ &= 0\end{aligned}$$

Question29

$$\text{If } \begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} > 0, \text{ then } abc >$$

AP EAPCET 2024 - 23th May Morning Shift

Options:

A. 1

B. -8

C. 8

D. 3

Answer: B

Solution:

$$\begin{aligned} \text{We have, } & \begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} > 0 \\ & = \{(abc - a) - 1(c - 1) + 1(1 - b)\} > 0 \\ & = (abc - a - c + 1 + 1 - b) > 0 \\ & = (abc - (a + b + c) + 2) > 0 \\ & = abc + 2 > (a + b + c) \end{aligned}$$

$$abc + 2 > 3(abc)^{\frac{1}{3}}$$

$$\left[\because Am > Gm \Rightarrow \frac{a+b+c}{3} > (a+b)^{1/3} \right]$$

$$\text{let } x = (abc)^{1/3} \Rightarrow abc = x^3$$

$$x^3 + 2 > 3x$$

$$x^3 - 3x + 2 > 0$$

$$(x - 1)^2(x + 2) > 0$$

$$(x + 2) > 0 = x > -2$$

$$1$$

$$= (abc)^{\frac{1}{3}} > -2 = (abc) > -8$$

Question30

If the system of equations

$$a_1x + b_1y + c_1z = 0, a_2x + b_2y + c_2z = 0 \text{ and}$$

$a_3x + b_3y + c_3z = 0$ has only trivial solution, then the rank of

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \text{ is}$$

AP EAPCET 2024 - 23th May Morning Shift

Options:

A. 2

B. 1

C. 3

D. 0

Answer: C

Solution:

The given system of equations are homogeneous and has only trivial solution, then

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

$\text{Rank}[A] = n$, where n is the order of square matrix and

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

$$\therefore \text{Rank}[A] = 3$$

Question31

$A = \begin{bmatrix} 0 & 1 & 2 \\ 2 & 3 & 0 \\ 4 & 0 & 3 \end{bmatrix}$ and B is a matrix such that $AB = BA$. If AB is not an identity matrix, then the matrix that can be taken as B is

AP EAPCET 2024 - 22th May Evening Shift

Options:

A. $\begin{bmatrix} -9 & -3 & 6 \\ -6 & 8 & -4 \\ 12 & -4 & -2 \end{bmatrix}$

B. $\begin{bmatrix} 9 & -3 & 6 \\ -6 & 8 & -4 \\ -12 & -4 & 2 \end{bmatrix}$

C. $\begin{bmatrix} 9 & -3 & -6 \\ -6 & 8 & -4 \\ -12 & 4 & -2 \end{bmatrix}$

D. $\begin{bmatrix} 9 & -3 & -6 \\ -6 & -8 & 4 \\ -12 & 4 & -2 \end{bmatrix}$

Answer: D

Solution:

Given,

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 2 & 3 & 0 \\ 4 & 0 & 3 \end{bmatrix}$$

and B is a matrix such that $AB = BA$. And AB is not an identity matrix.

Let

$$B = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$$

Now, $AB = BA$

$$\begin{bmatrix} 0 & 1 & 2 \\ 2 & 3 & 0 \\ 4 & 0 & 3 \end{bmatrix} \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 \\ 2 & 3 & 0 \\ 4 & 0 & 3 \end{bmatrix}$$
$$\begin{bmatrix} b_1 + 2c_1 & b_2 + 2c_2 & b_3 + 2c_3 \\ 2a_1 + 3b_1 & 2a_2 + 3b_2 & 2a_3 + 3b_3 \\ 4a_1 + 3c_1 & 4a_2 + 3c_2 & 4a_3 + 3c_3 \end{bmatrix} = \begin{bmatrix} 2a_2 + 4a_3 & a_1 + 3a_2 & 2a_1 + 3a_3 \\ 2b_2 + 4b_3 & b_1 + 3b_2 & 2b_1 + 3b_3 \\ 2c_2 + 4c_3 & c_1 + 3c_2 & 2c_1 + 3c_3 \end{bmatrix}$$

On comparing two matrix and solve for $a_1, a_2, a_3, b_1, b_2, b_3, c_1, c_2$ and c_3 , we get

$$a_1 = 9, a_2 = -3 \text{ and } a_3 = -6$$

$$b_1 = -6, b_2 = -8 \text{ and } b_3 = 4$$

$$c_1 = -12, c_2 = 4 \text{ and } c_3 = -2$$

$$\text{So, } B = \begin{bmatrix} 9 & -3 & -6 \\ -6 & -8 & 4 \\ -12 & 4 & -2 \end{bmatrix}$$

Question32

If α, β and γ ($\alpha < \beta < \gamma$) are the values of x such that

$\begin{bmatrix} x-2 & 0 & 1 \\ 1 & x+3 & 2 \\ 2 & 0 & 2x-1 \end{bmatrix}$ is a singular matrix, then $2\alpha + 3\beta + 4\gamma$ is equal to

AP EAPCET 2024 - 22th May Evening Shift

Options:

A. 4

B. 0

C. 1

D. 2

Answer: A

Solution:

Let $A = \begin{bmatrix} x-2 & 0 & 1 \\ 1 & x+3 & 2 \\ 2 & 0 & 2x-1 \end{bmatrix}$ is a singular matrix.

So, $|A| = 0$

$$\begin{vmatrix} x-2 & 0 & 1 \\ 1 & x+3 & 2 \\ 2 & 0 & 2x-1 \end{vmatrix} = 0$$

$$\begin{aligned} \Rightarrow (x-2)[(x+3)(2x-1) - 0] - 0 \\ \Rightarrow (x-2)(x+3)(2x-1) - 2(x+3) &= 0 \\ \Rightarrow (x+3)[(x-2)(2x-1) - 2] &= 0 \\ \Rightarrow (x+3)[2x^2 - x - 4x + 2 - 2] &= 0 \\ \Rightarrow (x+3)[2x^2 - 5x] &= 0 \\ \Rightarrow x(x+3)(2x-5) &= 0 \\ \Rightarrow x = -3, 0, 5/2 \end{aligned}$$

So, $\alpha = -3, \beta = 0$ and $\gamma = 5/2$

[given $\alpha < \beta < \gamma$]

Now,

$$\begin{aligned} 2\alpha + 3\beta + 4\gamma &= 2 \times (-3) + 3 \times 0 + 4 \times \frac{5}{2} \\ &= -6 + 10 = 4 \end{aligned}$$

Question33

The system of linear equations $x + 2y + z = -3, 3x + 3y - 2z = -1$ and $2x + 7y + 7z = -4$ has

AP EAPCET 2024 - 22th May Evening Shift

Options:

A. infinite number of solutions

B. no solution

C. unique solution

D. finite number of solutions

Answer: B

Solution:

To determine the nature of the solutions for the system of linear equations:

$$\begin{aligned}x + 2y + z &= -3, \\3x + 3y - 2z &= -1, \\2x + 7y + 7z &= -4,\end{aligned}$$

we begin by calculating the determinant Δ of the coefficient matrix:

$$\Delta = \begin{vmatrix} 1 & 2 & 1 \\ 3 & 3 & -2 \\ 2 & 7 & 7 \end{vmatrix}$$

Calculating this determinant:

$$\begin{aligned}\Delta &= 1(21 + 14) - 2(21 + 4) + 1(21 - 6) \\&= 1 \times 35 - 2 \times 25 + 1 \times 15 \\&= 35 - 50 + 15 = 0\end{aligned}$$

Since $\Delta = 0$, the system of equations is either inconsistent (no solution) or dependent (infinitely many solutions). To resolve this, we check Δ_x , the determinant of the matrix when the first column is replaced by the constants from the right side of the equations:

$$\Delta_x = \begin{vmatrix} -3 & 2 & 1 \\ -1 & 3 & -2 \\ -4 & 7 & 7 \end{vmatrix}$$

Calculating Δ_x :

$$\begin{aligned}\Delta_x &= (-3)(35) - 2(-15) + 1(5) \\&= -105 + 30 + 5 = -70\end{aligned}$$

Since $\Delta = 0$ and $\Delta_x \neq 0$, the system has no solution. Hence, the equations are inconsistent.

Question34

If the set of equations $x + 2y + 3z = 6$, $x + 3y + 5z = 9$, $2x + 5y + az = b$ has unique solution, then

AP EAPCET 2024 - 22th May Morning Shift

Options:

A. $a = 8, b = 15$

B. $a \neq 8, b \in R$

C. $a = 8, b \neq 15$

D. $a \neq 15, b = 8$

Answer: B

Solution:

To determine the condition for a and b so, that the set of equations

$$x + 2y + 3z = 6$$

$$x + 3y + 5z = 9$$

$2x + 5y + az = b$ has a unique solution. We need to ensure that the coefficient matrix has a non zero determinant.

$$\text{Let } A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 2 & 5 & a \end{pmatrix}$$

Then, $\det(A)$

$$= 1(3a - 25) - 2(a - 10) + 3(5 - 6)$$

$$= 1(3a - 25) - 2(a - 10) + 3(-1)$$

$$= 3a - 25 - 2a + 20 - 3$$

$$= a - 8$$

For system to have a unique solution. $\det(A)$ must be non-zero.

$$\text{i.e. } a - 8 \neq 0$$

Thus, $a \neq 8$

So, b can be other number which should be a non-zero real number and therefore b must be a real number.

Thus, $a \neq 8, b \in R$.

Question35

If P and Q are two 3×3 matrices such that $|PQ| = 1$ and $|P| = 9$, then the determinant of adjoint of the matrix P . $\text{adj } 3Q$ is

AP EAPCET 2024 - 22th May Morning Shift

Options:

A. 9^4

B. $\frac{1}{9^4}$

C. 9^2

D. $\frac{1}{9^2}$

Answer: A

Solution:

Given, $|PQ| = 1$ and $|P| = 9$.

For any $n \times n$ matrix A , the determinant of $\text{adj}(A)$ is $|A|^{n-1}$, where n is the dimension of the square matrix.

If $B = kA$, where k is a scalar, then $|B| = k^n|A|$ for an $n \times n$ matrix A .

Also, $|PQ| = |P||Q|$.

Utilizing these properties, for a 3×3 matrix we have:

$$|3Q| = 3^3|Q| = 27|Q|.$$

Now, considering the matrix $\text{adj}(3Q)$:

$$\text{Det}(\text{adj}(3Q)) = (27|Q|)^2.$$

Since $|PQ| = 1$ and $|P| = 9$, we have:

$$\begin{aligned} 1 &= 9 \cdot |Q|, \\ |Q| &= \frac{1}{9}. \end{aligned}$$

Finally, compute $\text{det}(\text{adj}(3Q))$:

$$\begin{aligned} \text{det}(\text{adj}(3Q)) &= (27|Q|)^2 \\ &= \left(27 \times \frac{1}{9}\right)^2 \\ &= 3^2 \\ &= 9. \end{aligned}$$

For $\text{adj}(P \cdot \text{adj}(3Q))$:

$$\begin{aligned} \text{adj}(P \cdot \text{adj}(3Q)) &= (9 \times 9)^{3-1} \\ &= (9^2)^2 \\ &= 9^4. \end{aligned}$$

Question36

If $A = \begin{bmatrix} a & 1 & 2 \\ 1 & 2 & b \\ c & 1 & 3 \end{bmatrix}$ and $\text{adj } A = \begin{bmatrix} 7 & -1 & -5 \\ -3 & 9 & 5 \\ 1 & -3 & 5 \end{bmatrix}$, then
 $a^2 + b^2 + c^2 =$

AP EAPCET 2024 - 22th May Morning Shift

Options:

A. 10

B. 14

C. 11

D. 29

Answer: A

Solution:

We know that for a square matrix the product $A \cdot \text{adj}(A) = \det(A) \cdot I$ where I is the identity matrix of the same order as A .

If A is a 3×3 matrix, then the product should be : $A \cdot \text{adj}(A) = \det(A) \cdot I_3$

Given matrix

$$A = \begin{bmatrix} a & 1 & 2 \\ 1 & 2 & b \\ c & 1 & 3 \end{bmatrix}$$

$$\text{adj}(A) = \begin{bmatrix} 7 & -1 & -5 \\ -3 & 9 & 5 \\ 1 & -3 & 5 \end{bmatrix}$$

Let's multiply A with $\text{adj}(A)$

$$A \cdot \text{adj}(A) = \begin{bmatrix} a & 1 & 2 \\ 1 & 2 & b \\ c & 1 & 3 \end{bmatrix} \begin{bmatrix} 7 & -1 & -5 \\ -3 & 9 & 5 \\ 1 & -3 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 7a - 1 & b + 1 & 7c \\ -a + 3 & -3b + 17 & -c \\ -5a + 15 & 5b + 5 & -5 + 20 \end{bmatrix}$$

Setting these elements equal to identity matrix adjusted by determinant

We get, $-a + 3 = 0$

$$\Rightarrow a = 3, b + 1 = 0 \Rightarrow b = -1$$

$$7c = 0 \Rightarrow c = 0$$

Thus, $a = 3, b = -1$ and $c = 0$

Hence, $a^2 + b^2 + c^2$

$$= 3^2 + (-1)^2 + 0 = 9 + 1 = 10$$

Question 37

If $3A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$ and $AA^T = I$, then $\frac{a}{b} + \frac{b}{a} =$

AP EAPCET 2024 - 21th May Evening Shift

Options:

A. $\frac{-5}{2}$

B. $\frac{13}{6}$

C. $-\frac{13}{6}$

D. $\frac{5}{2}$

Answer: D

Solution:

Given that $3A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$

$$\Rightarrow A = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ \frac{a}{3} & \frac{2}{3} & \frac{b}{3} \end{bmatrix}$$

$$\therefore A^T = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{a}{3} \\ \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{b}{3} \end{bmatrix}$$

$$\text{Now, } AA^T = I \Rightarrow \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ \frac{a}{3} & \frac{2}{3} & \frac{b}{3} \end{bmatrix} \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{a}{3} \\ \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{b}{3} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \frac{1}{9} + \frac{4}{9} + \frac{4}{9} & \frac{2}{9} + \frac{2}{9} - \frac{4}{9} & \frac{a}{9} + \frac{4}{9} + \frac{2b}{9} \\ \frac{2}{9} + \frac{2}{9} - \frac{4}{9} & \frac{4}{9} + \frac{1}{9} + \frac{4}{9} & \frac{2a}{9} + \frac{2}{9} - \frac{2b}{9} \\ \frac{a}{9} + \frac{4}{9} + \frac{2b}{9} & \frac{2a}{9} + \frac{2}{9} - \frac{2b}{9} & \frac{a^2}{9} + \frac{4}{9} + \frac{b^2}{9} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & \frac{a+4+2b}{9} \\ \frac{a+4+2b}{9} & \frac{2a+2-2b}{9} & \frac{a^2+4+b^2}{9} \\ \frac{a+4+2b}{9} & \frac{2a+2-2b}{9} & \frac{a^2+4+b^2}{9} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore a^2 + b^2 + 4 = 9 \quad \dots (i)$$

$$\Rightarrow a^2 + b^2 = 5$$

$$\text{and } 2a + 2 - 2b = 0$$

$$\Rightarrow b = a + 1 \quad \dots (ii)$$

$$\text{and } a + 4 + 2b = 0$$

$$\Rightarrow a + 2b = -4 \quad \dots (iii)$$

On solving Eqs. (ii) and (iii), we get

$$a + 2b = -4$$

$$\Rightarrow a + 2(a + 1) = -4$$

$$\Rightarrow a + 2a + 2 = -4$$

$$\Rightarrow 3a = -6$$

$$\Rightarrow a = -2 \text{ and } b = -2 + 1$$

$$\Rightarrow b = -1$$

$$\text{Now, } \frac{a}{b} + \frac{b}{a} = \frac{-2}{-1} + \frac{-1}{-2} = 2 + \frac{1}{2} = \frac{5}{2}$$

Question38

$$\begin{vmatrix} a + b + 2c & a & b \\ c & b + c + 2a & b \\ c & a & c + a + 2b \end{vmatrix} =$$

AP EAPCET 2024 - 21th May Evening Shift

Options:

A. $(a + b + c)^3$

B. $2(a + b + c)^3$

C. $3(a + b + c)^3$

D. $(a + b + c)$

Answer: B

Solution:

We have,

$$\begin{vmatrix} a+b+2c & a & b \\ c & b+c+2a & b \\ c & a & c+a+2b \end{vmatrix}$$

Applying $C_1 \rightarrow C_1 + C_2 + C_3$

$$= \begin{vmatrix} 2(a+b+c) & a & b \\ 2(b+c+a) & b+c+2a & b \\ 2(c+a+b) & a & c+a+2b \end{vmatrix}$$

$$= 2(a+b+c) \begin{vmatrix} 1 & a & b \\ 1 & b+c+2a & b \\ 1 & a & c+a+2b \end{vmatrix}$$

Applying $R_1 \rightarrow R_1 - R_2$ and $R_2 \rightarrow R_2 - R_3$

$$= 2(a+b+c) \begin{vmatrix} 0 & -(a+b+c) & 0 \\ 0 & a+b+c & -(a+b+c) \\ 1 & a & c+a+2b \end{vmatrix}$$

$$= 2(a+b+c)^3 \begin{vmatrix} 0 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & a & c+a+2b \end{vmatrix}$$

$$= 2(a+b+c)^3 [1(1-0)] = 2(a+b+c)^3$$

Question39

Assertion (A) : If B is a 3×3 matrix and $|B| = 6$, then $|\text{adj}(B)| = 36$

Reason (R) : If B is a square matrix of order n , then $|\text{adj}(B)| = |B|^n$

AP EAPCET 2024 - 21th May Evening Shift

Options:

- A. Both (A) and (R) are true and (R) is the correct explanation of (A).
- B. Both (A) and (R) are true but (R) is not the correct explanation of (A).
- C. (A) is true but (R) is false.
- D. (A) is false but (R) is true.

Answer: C

Solution:

To clarify:

When we have a matrix B of order 3 with a determinant $|B| = 6$, the formula to find the determinant of its adjugate matrix, $|\text{adj}(B)|$, is given by:

$$|\text{adj}(B)| = |B|^{n-1}$$

where n is the order of the matrix B .

In this case, since $n = 3$, we have:

$$|\text{adj}(B)| = 6^{3-1} = 6^2 = 36$$

This confirms that the assertion is true.

However, the reason provided, $|\text{adj}(B)| = |B|^n$, is incorrect because it states that the determinant of the adjugate matrix is raised to the power of n rather than $n - 1$. The correct relationship involves raising the determinant of B to the power of $n - 1$.

Thus, the assertion is true, but the given reason is false.

Question40

If $A = \begin{vmatrix} 2 & 3 & 4 \\ 1 & k & 2 \\ 4 & 1 & 5 \end{vmatrix}$ is singular matrix, then the quadratic equation having the roots k and $\frac{1}{k}$ is

AP EAPCET 2024 - 21th May Morning Shift

Options:

A. $6x^2 + 13x + 6 = 0$

B. $12x^2 - 25x + 12 = 0$

C. $6x^2 - 13x + 6 = 0$

D. $2x^2 - 5x + 2 = 0$

Answer: C

Solution:

Given the matrix $A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & k & 2 \\ 4 & 1 & 5 \end{bmatrix}$, which is singular, we need to find the quadratic equation with roots k and $\frac{1}{k}$.

Since A is singular, its determinant is zero:

$$|A| = \begin{vmatrix} 2 & 3 & 4 \\ 1 & k & 2 \\ 4 & 1 & 5 \end{vmatrix} = 2(5k - 2) - 3(5 - 8) + 4(1 - 4k) = 0$$

Expanding this:

$$2(5k - 2) = 10k - 4$$

$$-3(5 - 8) = -3 \cdot 3 = -9$$

$$4(1 - 4k) = 4 - 16k$$

Putting it all together, we get:

$$10k - 4 - 9 + 4 - 16k = 0$$

Simplifying:

$$-6k + 9 = 0$$

Solving for k :

$$k = \frac{3}{2}$$

Thus, the roots of the quadratic equation are $\frac{3}{2}$ and $\frac{2}{3}$. Using the standard form of a quadratic equation $x^2 - (\alpha + \beta)x + \alpha\beta = 0$ where the roots are $\alpha = \frac{3}{2}$ and $\beta = \frac{2}{3}$, we calculate:

$$\alpha + \beta = \frac{3}{2} + \frac{2}{3} = \frac{9}{6} + \frac{4}{6} = \frac{13}{6}$$

$$\alpha\beta = \frac{3}{2} \cdot \frac{2}{3} = 1$$

Substitute these into the quadratic equation:

$$x^2 - \left(\frac{13}{6}\right)x + 1 = 0$$

To eliminate fractions, multiply through by 6:

$$6x^2 - 13x + 6 = 0$$

This is the quadratic equation with the specified roots.

Question41

Let A be a 4×4 matrix and P be is adjoint matrix, If $|P| = \left| \frac{A}{2} \right|$ then $|A^{-1}|$

AP EAPCET 2024 - 21th May Morning Shift

Options:

A. $\pm \frac{1}{4}$

B. ± 8

C. ± 2

D. ± 4

Answer: D

Solution:

Given that $|P| = \left|\frac{A}{2}\right|$, we start by using the property of the adjoint matrix, where $|\text{adj}(A)| = |A|^{n-1}$ for an $n \times n$ matrix. Here, since A is a 4×4 matrix, $|\text{adj}(A)| = |A|^3$.

Substituting the given condition, we have:

$$|A|^3 = \left(\frac{1}{2}\right)^4 |A|$$

Simplifying, this becomes:

$$|A|^3 = \frac{1}{16} |A|$$

Assuming $|A| \neq 0$, divide both sides by $|A|$:

$$|A|^2 = \frac{1}{16}$$

Taking the square root on both sides gives:

$$|A| = \pm \frac{1}{4}$$

To find $|A^{-1}|$, we use the property $|A^{-1}| = \frac{1}{|A|}$:

$$|A^{-1}| = \pm 4$$

Therefore, the absolute determinant of the inverse of A is ± 4 .

Question42

The system $x + 2y + 3z = 4$, $4x + 5y + 3z = 5$, $3x + 4y + 3z = \lambda$ is consistent and $3\lambda = n + 100$, then $n =$

AP EAPCET 2024 - 21th May Morning Shift

Options:

A. -42

B. -86

C. 16

D. -24

Answer: B

Solution:

We have,

$$x + 2y + 3z = 4$$

$$4x + 5y + 3z = 5$$

$$3x + 4y + 3z = \lambda$$

Write the equation in matrix form $[A_1^1 B]$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 4 & 5 & 3 & 5 \\ 3 & 4 & 3 & \lambda \end{array} \right]$$

$$R_2 \rightarrow R_2 - 4R_1 \text{ and } R_3 \rightarrow R_3 - 3R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & -3 & -9 & -11 \\ 0 & -2 & -6 & \lambda - 12 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & -3 & -9 & -11 \\ 0 & 0 & 0 & 3\lambda - 14 \end{array} \right]$$

$$R_3 \leftrightarrow 3R_3 - 2R_2$$

For consistency $3\lambda - 14 = 0 \Rightarrow 3\lambda = 14$

Given, $3\lambda = n + 100$

Therefore, $n = 14 - 100$

$$n = -86$$

So, the value of n is -86 .

Question43

$$\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ 1 & 1 & 1 \end{vmatrix} \text{ is not equal to}$$

AP EAPCET 2024 - 20th May Evening Shift

Options:

A. $\begin{vmatrix} a+1 & b+1 & c+1 \\ a^2+1 & b^2+1 & c^2+1 \\ 1 & 1 & 1 \end{vmatrix}$

B. $\begin{vmatrix} a-b & b-c & c \\ a^2-b^2 & b^2-c^2 & c^2 \\ 0 & 0 & 1 \end{vmatrix}$

$$C. \begin{vmatrix} a(a+1) & b(b+1) & c(c+1) \\ a+1 & b+1 & c+1 \\ -1 & -1 & -1 \end{vmatrix}$$

$$D. \begin{vmatrix} a+b & b+c & c+a \\ a^2+b^2 & b^2+c^2 & c^2+a^2 \\ 2 & 2 & 2 \end{vmatrix}$$

Answer: D

Solution:

$$\text{Let } \Delta = \begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ 1 & 1 & 1 \end{vmatrix} \dots \text{(i)}$$

Applying, $R_1 \rightarrow R_1 + R_3, R_2 \rightarrow R_2 + R_3$ we get

$$\Delta = \begin{vmatrix} a+1 & b+1 & 0+1 \\ a^2+1 & b^2+1 & c^2+1 \\ 1 & 1 & 1 \end{vmatrix}$$

Applying $C_1 \rightarrow C_1 - C_2$ and $C_2 \rightarrow C_2 - C_3$, we get

$$\begin{aligned} \Delta &= \begin{vmatrix} a-b & b-c & c+1 \\ a^2-b^2 & b^2-c^2 & c^2+1 \\ 0 & 0 & 1 \end{vmatrix} \\ &= 1 [(a-b)(b^2-c^2) - (b-c)(a^2-b^2)] \\ &= 1 [ab^2 - ac^2 - b^3 + bc^2 - a^2b + b^3 + a^2c - b^2c] \\ &= ab^2 - a^2b - ac^2 + a^2c - b^2c + bc^2 \end{aligned}$$

Thus, option (a) and (b) are correct. Now, option (c),

$$\begin{vmatrix} a(a+1) & b(b+1) & c(c+1) \\ a+1 & b+1 & c+1 \\ -1 & -1 & -1 \end{vmatrix}$$

Applying, $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$

$$\begin{aligned} &= \begin{vmatrix} a(a+1) & b^2+b-a^2-a & c^2+c-a^2-a \\ a+1 & b-a & c-a \\ -1 & 0 & 0 \end{vmatrix} \\ &= -1 [(b^2+b-a^2-a)(c-a) - (c^2+c-a^2-a)(b-a)] \\ &= (c^2+c-a^2-a)(b-a) - (b^2+b-a^2-a)(c-a) \\ &= c^2b - ac^2 + bc - ac - a^2b + a^3 - ab + a^2 - b^2c + ab^2 - bc + ab + a^2c - a^3 + ac - a^2 \end{aligned}$$

Hence, option (d) is correct required answer.

Question44

Let A, B, C, D and E be $n \times n$ matrices each with non-zero determinant. If $ABCDE = I$, then $C^{-1} =$

AP EAPCET 2024 - 20th May Evening Shift

Options:

A. $E^{-1}D^{-1}B^{-1}A^{-1}$

B. $DEAB$

C. $A^{-1}B^{-1}D^{-1}E^{-1}$

D. $ABDE$

Answer: B

Solution:

Given, $ABCDE = I$

$$\Rightarrow A^{-1}(ABCDE)E^{-1} = A^{-1}IE^{-1}$$

($\because$ Pre multiply by A^{-1} and post multiply by E^{-1} , respectively)

$$\Rightarrow IBCDI = A^{-1}E^{-1}$$

$$\Rightarrow BCD = A^{-1}E^{-1}$$

Now, pre multiply by B^{-1} and post multiply by D^{-1} , we get

$$\Rightarrow B^{-1}(BCD)D^{-1} = B^{-1}A^{-1}E^{-1}D^{-1}$$

$$\Rightarrow B^{-1}BCDD^{-1} = B^{-1}A^{-1}E^{-1}D^{-1}$$

$$\Rightarrow ICI = B^{-1}A^{-1}E^{-1}D^{-1}$$

$$\Rightarrow C = B^{-1}A^{-1}E^{-1}D^{-1}$$

$$\begin{aligned}\therefore C^{-1} &= (B^{-1}A^{-1}E^{-1}D^{-1})^{-1} \\ &= (D^{-1})^{-1}(E^{-1})^{-1}(A^{-1})^{-1}(B^{-1})^{-1} \\ &\quad \text{(using reversal law)}\end{aligned}$$

$$= DEAB$$

$$\therefore C^{-1} = DEAB$$

Question45

If $A = [a_{ij}]$, $1 \leq i, j \leq n$ with $n \geq 2$ and $a_{ij} = i + j$ is a matrix, then the rank of A is

AP EAPCET 2024 - 20th May Evening Shift

Options:

A. 0

B. 1

C. 2

D. 4

Answer: C

Solution:

Given, $A = [a_{ij}]$, $1 \leq i, j \leq n$ with $n \geq 2$ and $a_{ij} = i + j$

$\therefore$ The matrix is

$$A = \begin{bmatrix} 2 & 3 & 4 & \cdots & n+1 \\ 3 & 4 & 5 & \cdots & n+2 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ n+1 & n+2 & n+3 & \cdots & n+n \end{bmatrix}$$
$$= \begin{bmatrix} 2 & 3 & \cdots & n+1 \\ 3 & 4 & \cdots & n+2 \\ 4 & 5 & \cdots & n+3 \\ \vdots & \vdots & \cdots & \vdots \\ n+1 & n+2 & \cdots & 2n \end{bmatrix}$$

Use elementary row operation to subtract 1st row from the i th row for $2 \leq i \leq n$, we get

$$A = \begin{bmatrix} 2 & 3 & 4 & \cdots & n+1 \\ 1 & 1 & 1 & \cdots & 1 \\ 2 & 2 & 2 & \cdots & 2 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ n-1 & n-1 & n-1 & \cdots & n-1 \end{bmatrix}$$

Applying, $R_3 \rightarrow R_3 - 2R_2$ and $R_4 \rightarrow R_4 - 3R_2$ and so on

$$A = \begin{bmatrix} 2 & 3 & 4 & \cdots & n+1 \\ 1 & 1 & 1 & \cdots & 1 \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$$

$\therefore$ Rank $A = 2$

Question 46

If $A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 3 \\ 3 & 2 & 4 \end{bmatrix}$, then $A^2 - 5A + 6I =$

AP EAPCET 2024 - 20th May Morning Shift

Options:

A. $\begin{bmatrix} 8 & 4 & 0 \\ 3 & 8 & 4 \\ 4 & 0 & 12 \end{bmatrix}$

B. $\begin{bmatrix} 8 & 4 & 0 \\ 3 & 6 & 4 \\ 4 & 0 & 14 \end{bmatrix}$

C. $\begin{bmatrix} 8 & 6 & 0 \\ 3 & 8 & 4 \\ 2 & 0 & 14 \end{bmatrix}$

D. $\begin{bmatrix} 8 & 4 & 0 \\ 3 & 8 & 4 \\ 4 & 0 & 14 \end{bmatrix}$

Answer: D

Solution:

Given the matrix A :

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 3 \\ 3 & 2 & 4 \end{bmatrix}$$

We are tasked to find $A^2 - 5A + 6I$.

Calculate A^2 :

$$A^2 = A \cdot A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 3 \\ 3 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 3 \\ 3 & 2 & 4 \end{bmatrix}$$

Perform the matrix multiplication:

$$A^2 = \begin{bmatrix} 1 \cdot 1 + 0 \cdot 2 + 2 \cdot 3 & 1 \cdot 0 + 0 \cdot 1 + 2 \cdot 2 & 1 \cdot 2 + 0 \cdot 3 + 2 \cdot 4 \\ 2 \cdot 1 + 1 \cdot 2 + 3 \cdot 3 & 2 \cdot 0 + 1 \cdot 1 + 3 \cdot 2 & 2 \cdot 2 + 1 \cdot 3 + 3 \cdot 4 \\ 3 \cdot 1 + 2 \cdot 2 + 4 \cdot 3 & 3 \cdot 0 + 2 \cdot 1 + 4 \cdot 2 & 3 \cdot 2 + 2 \cdot 3 + 4 \cdot 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1+0+6 & 0+0+4 & 2+0+8 \\ 2+2+9 & 0+1+6 & 4+3+12 \\ 3+4+12 & 0+2+8 & 6+6+16 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & 4 & 10 \\ 13 & 7 & 19 \\ 19 & 10 & 28 \end{bmatrix}$$

Calculate $5A$:

$$5A = 5 \cdot \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 3 \\ 3 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 10 \\ 10 & 5 & 15 \\ 15 & 10 & 20 \end{bmatrix}$$

Calculate $6I$:

$$6I = 6 \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

Combine the results to find $A^2 - 5A + 6I$:

$$A^2 - 5A + 6I = \begin{bmatrix} 7 & 4 & 10 \\ 13 & 7 & 19 \\ 19 & 10 & 28 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 10 \\ 10 & 5 & 15 \\ 15 & 10 & 20 \end{bmatrix} + \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 7-5+6 & 4-0+0 & 10-10+0 \\ 13-10+0 & 7-5+6 & 19-15+0 \\ 19-15+0 & 10-10+0 & 28-20+6 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 4 & 0 \\ 3 & 8 & 4 \\ 4 & 0 & 14 \end{bmatrix}$$

Thus, the resulting matrix is:

$$\begin{bmatrix} 8 & 4 & 0 \\ 3 & 8 & 4 \\ 4 & 0 & 14 \end{bmatrix}$$

Question47

Sum of the positive roots of the equation $\begin{vmatrix} x^2 + 2x & x + 2 & 1 \\ 2x + 1 & x - 1 & 1 \\ x + 2 & -1 & 1 \end{vmatrix} = 0$ **is**

AP EAPCET 2024 - 20th May Morning Shift

Options:

A. $\frac{1+\sqrt{13}}{2}$

B. 1

C. $\frac{\sqrt{13}-1}{2}$

D. 3

Answer: A

Solution:

$$\text{We have, } \begin{vmatrix} x^2 + 2x & x + 2 & 1 \\ 2x + 1 & x - 1 & 1 \\ x + 2 & -1 & 1 \end{vmatrix} = 0$$

$$\begin{aligned} &\Rightarrow (x^2 + 2x) \begin{vmatrix} x - 1 & 1 \\ -1 & 1 \end{vmatrix} - (x + 2) \begin{vmatrix} 2x + 1 & 1 \\ x + 2 & 1 \end{vmatrix} + 1 \\ &\Rightarrow (x^2 + 2x)(x - 1 + 1) - (x + 2)(2x + 1 - x - 2) \\ &\quad + 1(-2x - 1 - x^2 - x + 2) = 0 \\ &\Rightarrow x(x^2 + 2x) - (x + 2)(x - 1) + 1 \\ &\quad (-x^2 - 3x + 1) = 0 \\ &\Rightarrow x^3 + 2x^2 - x^2 - x + 2 - x^2 - 3x + 1 = 0 \\ &\Rightarrow x^3 - 4x + 3 = 0 \end{aligned}$$

On putting, $x = 1$, we get

$$(1)^3 - 4 \times 1 + 3 = 0$$

Thus, $x = 1$ is a root of the given equation.

$$\text{Thus, } (x - 1)(x^2 + x - 3) = 0$$

$$\text{Solve } x^2 + x - 3 = 0$$

$$\begin{aligned} x &= \frac{-1 \pm \sqrt{1^2 - 4 \times 1 \times (-3)}}{2 \times 1} \\ &= \frac{-1 \pm \sqrt{1 + 12}}{2} = \frac{-1 \pm \sqrt{13}}{2} \end{aligned}$$

Hence, positive roots are 1 and $\frac{-1+\sqrt{13}}{2}$

$$\begin{aligned} \text{Sum of positive roots} &= 1 + \left(\frac{-1+\sqrt{13}}{2} \right) \\ &= \frac{1+\sqrt{13}}{2} \end{aligned}$$

Question48

If the solution of the system of simultaneous linear equations $x + y - z = 6$, $3x + 2y - z = 5$ and $2x - y - 2z + 3 = 0$ is $x = \alpha$, $y = \beta$, $z = \gamma$, then $\alpha + \beta =$

AP EAPCET 2024 - 20th May Morning Shift

Options:

A. -7

B. 2

C. 1

D. -2

Answer: B

Solution:

Given system of equation

$$x + y - z = 6$$

$$3x + 2y - z = 5$$

$$2x - y - 2z = -3$$

We can represent this as

$$A \cdot X = B$$

Where

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 3 & 2 & -1 \\ 2 & -1 & -2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 6 \\ 5 \\ -3 \end{bmatrix}$$

$$\text{Now, } [A/B] = \begin{bmatrix} 1 & 1 & -1 \\ 3 & 2 & -1 \\ 2 & -1 & -2 \end{bmatrix} \left\{ \begin{bmatrix} 6 \\ 5 \\ -3 \end{bmatrix} \right.$$

$$R_2 \rightarrow R_2 - 3R_1 = \begin{bmatrix} 1 & 1 & -1 \\ 0 & -1 & 2 \\ 2 & -1 & -2 \end{bmatrix} \left\{ \begin{bmatrix} 6 \\ -13 \\ -3 \end{bmatrix} \right.$$

$$R_3 \rightarrow R_3 - 2R_1 = \begin{bmatrix} 1 & 1 & -1 \\ 0 & -1 & 2 \\ 0 & -3 & 0 \end{bmatrix} \left\{ \begin{bmatrix} 6 \\ -13 \\ -15 \end{bmatrix} \right.$$

$$R_2 \rightarrow -R_2 = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -2 \\ 0 & -3 & 0 \end{bmatrix} \left\{ \begin{bmatrix} 6 \\ 13 \\ -15 \end{bmatrix} \right.$$

$$R_3 \rightarrow R_3 + 3R_2 = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & -6 \end{bmatrix} \left\{ \begin{bmatrix} 6 \\ 13 \\ 24 \end{bmatrix} \right.$$

$$R_3 \rightarrow \frac{R_3}{-6}$$
$$= \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \left\{ \begin{bmatrix} 6 \\ 13 \\ -4 \end{bmatrix} \right.$$

$$R_2 \rightarrow R_2 + 2R_3 = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 5 \\ -4 \end{bmatrix}$$

Now, we get

$$x + y - z = 6$$

$$y = 5$$

$$z = -4$$

$$x + 5 + 4 = 6$$

$$x + 9 = 6$$

$$x = -3$$

Thus, $x = \alpha = -3$

$$y = \beta = 5$$

$$z = \gamma = -4$$

then, $\alpha + \beta = -3 + 5 = 2$

Question49

$$\begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} =$$

AP EAPCET 2024 - 19th May Evening Shift

Options:

A. $(a - b)(b - c)(c - a)(a + b + c)$

B. $(a - b)(b - c)(c - a)$

C. $(a - b)(b - c)(a - c)(ab + bc + ca)$

D. $(a - b)(b - c)(c - a)(ab + bc + ca)$

Answer: D

Solution:

We have, $\begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix}$

$$\begin{aligned}
& C_2 \rightarrow C_2 - C_1 \text{ and } C_3 \rightarrow C_3 - C_1 \\
& = \begin{vmatrix} 1 & 0 & 0 \\ a^2 & b^2 - a^2 & c^2 - a^2 \\ a^3 & b^3 - a^3 & c^3 - a^3 \end{vmatrix} \\
& = (b-a)(c-a) \\
& \begin{vmatrix} 1 & 0 & 0 \\ a^2 & (b+a) & (c+a) \\ a^3 & (b^2 + a^2 + ab) & (c^2 + a^2 + ac) \end{vmatrix} \\
& = (b-a)(c-a) \\
& \begin{vmatrix} b+a & c+a \\ b^2 + a^2 + ab & c^2 + a^2 + ac \end{vmatrix} \\
& = [(b-a)(c-a) [bc^2 + a^2b + abc + ac^2 + a^3 \\
& + a^2c - b^2c - a^3 - a^2b - a^2c - abc - ab^2] \\
& = (b-a)(c-a) [bc^2 + ac^2 - b^2c - ab^2] \\
& = (b-a)(c-a)[bc(c-b) \\
& + a(c-b)(c+b)] \\
& = (a-b)(b-c)(c-a)(ab+bc+ca)
\end{aligned}$$

Question50

If $A = \begin{bmatrix} 1 & 2 \\ -2 & -5 \end{bmatrix}$ and $\alpha A^2 + \beta A = 2I$ for some $\alpha, \beta \in R$, then $\alpha + \beta =$

AP EAPCET 2024 - 19th May Evening Shift

Options:

- A. 7
- B. 10
- C. 12
- D. 5

Answer: B

Solution:

To solve the given equation $\alpha A^2 + \beta A = 2I$ where $A = \begin{bmatrix} 1 & 2 \\ -2 & -5 \end{bmatrix}$, we first need to compute A^2 :

$$A^2 = \begin{bmatrix} 1 & 2 \\ -2 & -5 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -5 \end{bmatrix}$$

Calculating the matrix multiplication:

$$A^2 = \begin{bmatrix} (1)(1) + (2)(-2) & (1)(2) + (2)(-5) \\ (-2)(1) + (-5)(-2) & (-2)(2) + (-5)(-5) \end{bmatrix} = \begin{bmatrix} -3 & -8 \\ 8 & 21 \end{bmatrix}$$

Next, substitute A^2 into the equation $\alpha A^2 + \beta A = 2I$:

$$\alpha \begin{bmatrix} -3 & -8 \\ 8 & 21 \end{bmatrix} + \beta \begin{bmatrix} 1 & 2 \\ -2 & -5 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

Break this down into:

$$\begin{bmatrix} -3\alpha & -8\alpha \\ 8\alpha & 21\alpha \end{bmatrix} + \begin{bmatrix} \beta & 2\beta \\ -2\beta & -5\beta \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

Equate the corresponding elements from both sides:

$$-3\alpha + \beta = 2$$

$$-8\alpha + 2\beta = 0$$

$$8\alpha - 2\beta = 0$$

$$21\alpha - 5\beta = 2$$

From the second equation $-8\alpha + 2\beta = 0$, we solve for β :

$$2\beta = 8\alpha \Rightarrow \beta = 4\alpha$$

Substitute $\beta = 4\alpha$ into the first equation:

$$-3\alpha + 4\alpha = 2 \Rightarrow \alpha = 2$$

Using $\beta = 4\alpha$, we find:

$$\beta = 4(2) = 8$$

Finally, calculate $\alpha + \beta$:

$$\alpha + \beta = 2 + 8 = 10$$

Question51

The system of equations $x + 2y + 3z = 6$, $x + 3y + 5z = 9$, $2x + 5y + az = 12$ has no solution when $a =$

AP EAPCET 2024 - 19th May Evening Shift

Options:

A. 5

B. 6

C. 7

D. 8

Answer: D

Solution:

To find the value of a that makes the system of equations have no solution, we need to ensure that the determinant of the coefficient matrix is zero. This will indicate that the equations are linearly dependent, leading to no unique solutions.

Given the system of equations:

$$\begin{aligned}x + 2y + 3z &= 6, \\x + 3y + 5z &= 9, \\2x + 5y + az &= 12,\end{aligned}$$

we consider the coefficient matrix:

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 2 & 5 & a \end{bmatrix}$$

Calculate the determinant of this matrix:

$$D = \begin{vmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 2 & 5 & a \end{vmatrix}$$

The determinant is calculated as follows:

$$\begin{aligned}D &= 1 \cdot (3a - 25) - 2 \cdot (a - 10) + 3 \cdot (5 - 6) \\&= 3a - 25 - 2a + 20 - 3 \\&= 3a - 2a - 25 + 20 - 3 \\&= a - 8.\end{aligned}$$

Setting the determinant $D = 0$ gives:

$$a - 8 = 0,$$

which simplifies to:

$$a = 8.$$

Therefore, the system of equations has no solution when $a = 8$.

Question52

If α, β, γ are the roots of $\begin{bmatrix} 1 & -x & -2 \\ -2 & 4 & -x \\ -2 & 1 & -x \end{bmatrix} = 0$, then $\alpha\beta + \beta\gamma + \gamma\alpha =$

AP EAPCET 2024 - 18th May Morning Shift

Options:

- A. 6
- B. 8
- C. 0
- D. -4

Answer: C

Solution:

We have.

$$\begin{vmatrix} 1-x & -2 & 1 \\ -2 & 4-x & -2 \\ 1 & -2 & 1-x \end{vmatrix} = 0$$

$R_1 \rightarrow R_1 + R_2 + R_3$ we get

$$\Rightarrow \begin{vmatrix} -x & -x & -x \\ -2 & 4-x & -2 \\ 1 & -2 & 1-x \end{vmatrix} = 0$$

$$\Rightarrow -x \begin{vmatrix} 1 & 1 & 1 \\ -2 & 4-x & -2 \\ 1 & -2 & 1-x \end{vmatrix} = 0$$

$C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$, we get

$$\Rightarrow -x \begin{vmatrix} 1 & 0 & 0 \\ -2 & 6-x & 0 \\ 1 & -3 & -x \end{vmatrix} = 0$$

$$\Rightarrow (-x)(-x)(6-x) = 0$$

$$x' - 6x^2 = 0$$

Sum of product of zero

$$= \frac{\text{Coefficient of } x}{\text{Coefficient of } x^3} = 0$$

Hence, $\alpha\beta + \beta\gamma + p^2 = 0$

Question53

If the determinant of a 3rd order matrix A is K , then the sum of the determinants of the matrices A^4 and $(A - A^4)$ is

AP EAPCET 2024 - 18th May Morning Shift

Options:

A. $2K$

B. 0

C. K^2

D. K

Answer: C

Solution:

We are given that the determinant of a 3rd order matrix A is K . Hence, $|A| = K$.

For a matrix A , the determinant properties used here are:

The determinant of the transpose of a matrix is the same as the determinant of the matrix itself. So, $|A^T| = |A| = K$.

The determinant of the product of a matrix with its transpose is $|AA^T| = |A||A^T| = K \times K = K^2$.

Now, we are asked to find the sum of the determinants of matrices A^4 and $(A - A^4)$.

Since the determinant of a product is the product of the determinants and given $|A| = K$, we have:

$$|A^4| = |A|^4 = K^4.$$

To find $|A - A^4|$, note that:

By matrix properties, $|(A - A^4)| =$ simplify or evaluate separately.

Given the expression $|(AA^T) + (A - A^T)|$ in the context, we simplify:

$$|AA^T| + |(A - A^4)| = K^2 + |A| - |A^T| = K^2 + K - K = K^2.$$

Thus the result for the sum of these determinants aligns with the property calculations stated, giving us the simplified determinant result of K^2 .

Question54

While solving a system of linear equations $AX = B$ using Cramer's rule with the usual notation if

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 2 & -1 & 2 \\ -1 & 1 & 5 \end{vmatrix}, \Delta_1 = \begin{vmatrix} 5 & 1 & 1 \\ 4 & -1 & 2 \\ 11 & 1 & 5 \end{vmatrix} \text{ and } X = \begin{bmatrix} \alpha \\ 2 \\ \beta \end{bmatrix}, \text{ then } \alpha^2 + \beta^2 =$$

Options:

- A. 9
- B. 13
- C. 5
- D. 25

Answer: C**Solution:**

We have,

$$\begin{aligned}
 \Delta &= \begin{vmatrix} 1 & 1 & 1 \\ 2 & -1 & 2 \\ -1 & 1 & 5 \end{vmatrix} \\
 &= 1(-5 - 2) - 1(10 + 2) + 1(2 - 1) \\
 &= -7 - 12 + 1 \\
 &= -18 \neq 0 \\
 \Delta_1 &= \begin{vmatrix} 5 & 1 & 1 \\ 4 & -1 & 2 \\ 11 & 1 & 5 \end{vmatrix} \\
 &= 5(-5 - 2) - 1(20 - 22) + 1(4 + 11) \\
 &= -35 + 2 + 15 \\
 &= -18 \\
 \Delta_3 &= \begin{vmatrix} 1 & 1 & 5 \\ 2 & -1 & 4 \\ -1 & 1 & 11 \end{vmatrix} \\
 &= 1(-11 - 4) - 1(22 + 4) + 5(2 - 1) \\
 &= -15 - 26 + 5 = -36 \\
 \therefore \alpha &= \frac{\Delta_1}{\Delta} = \frac{-18}{-18} = 1, \\
 \therefore \beta &= \frac{\Delta_3}{\Delta} = \frac{-36}{-18} = 2
 \end{aligned}$$

Hence, $\alpha^2 + \beta^2 = 1^2 + 2^2 = 5$.**Question55**

If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, then AA^T is a

AP EAPCET 2022 - 5th July Morning Shift

Options:

- A. symmetric matrix
- B. skew-symmetric matrix
- C. singular matrix
- D. inverse of A

Answer: A

Solution:

Here,

$$A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 3 & 2 & 0 \\ -3 & -3 & -1 \\ 4 & 4 & 1 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 2 & 0 \\ -3 & -3 & -1 \\ 4 & 4 & 1 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 9+9+16 & 6+9+16 & 0+3+4 \\ 6+9+16 & 4+9+16 & 0+3+4 \\ 0+3+4 & 0+3+4 & 0+1+1 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 34 & 31 & 7 \\ 31 & 29 & 7 \\ 7 & 7 & 2 \end{bmatrix}$$

$$(AA^T)^T = \begin{bmatrix} 34 & 31 & 7 \\ 31 & 29 & 7 \\ 7 & 7 & 2 \end{bmatrix}$$

So, AA^T is a symmetric matrix.

Question 56

If $AX = D$ represents the system of simultaneous linear equations $x + y + z = 6$, $5x - y + 2z = 3$ and $2x + y - z = -5$, then $(\text{Adj } A) D =$

Options:

A. $\begin{bmatrix} 8 \\ -16 \\ 40 \end{bmatrix}$

B. $\begin{bmatrix} 32 \\ 64 \\ -160 \end{bmatrix}$

C. $\begin{bmatrix} -16 \\ 32 \\ 80 \end{bmatrix}$

D. $\begin{bmatrix} 12 \\ 24 \\ 60 \end{bmatrix}$

Answer: C**Solution:**

The given system of equation is $5x - y + 2z = 3$, $x + y + z = 6$ and $2x + y - z = -5$ In matrix form, it may be represented as where

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 5 & -1 & 2 \\ 2 & 1 & -1 \end{bmatrix}, x = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } D = \begin{bmatrix} 6 \\ 3 \\ -5 \end{bmatrix}$$

Thus, A is a non-singular matrix, so A^{-1} exist.

$$C_{11} = (-1)^{1+1} \begin{vmatrix} -1 & 2 \\ 1 & -1 \end{vmatrix} = -1$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} 5 & 2 \\ 2 & -1 \end{vmatrix} = 9$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} 5 & -1 \\ 2 & 1 \end{vmatrix} = 7$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = 2$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 1 \\ 2 & -1 \end{vmatrix} = -3$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = 1$$

$$C_{31} = (-1)^{3+1} \begin{vmatrix} 1 & 1 \\ -1 & 2 \end{vmatrix} = 3$$

$$C_{32} = (-1)^{3+2} \begin{vmatrix} 1 & 1 \\ 5 & 2 \end{vmatrix} = 3$$

$$C_{33} = (-1)^{3+3} \begin{vmatrix} 1 & 1 \\ 5 & -1 \end{vmatrix} = -6$$

$$\begin{aligned}\text{Then, } \text{adj}(A) &= \begin{bmatrix} -1 & 9 & 7 \\ 2 & -3 & 1 \\ 3 & 3 & -6 \end{bmatrix}^T = \begin{bmatrix} -1 & 2 & 3 \\ 9 & -3 & 3 \\ 7 & 1 & -6 \end{bmatrix} \\ (\text{adj } A) \cdot D &= \begin{bmatrix} -1 & 2 & 3 \\ 9 & -3 & 3 \\ 7 & 1 & -6 \end{bmatrix} \begin{bmatrix} 6 \\ 3 \\ -5 \end{bmatrix} \\ &= \begin{bmatrix} -6 + 6 - 15 \\ 54 - 9 - 15 \\ 42 + 3 + 30 \end{bmatrix} = \begin{bmatrix} -15 \\ 30 \\ 75 \end{bmatrix}\end{aligned}$$

Question 57

If $A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$, then $\det(A^6 + B^6) =$

AP EAPCET 2022 - 5th July Morning Shift

Options:

A. -68

B.

- 212

C. 665

D. 720

Answer: B

Solution:

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 1+0 & 0+0 \\ 2+2 & 0+1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$$

$$\Rightarrow A^3 = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \quad [A^2 \cdot A = A^3]$$

$$\Rightarrow A^3 = \begin{bmatrix} 1+0 & 0+0 \\ 4+2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 6 & 1 \end{bmatrix}$$

Similarly, $A^6 = \begin{bmatrix} 1 & 0 \\ 12 & 1 \end{bmatrix} \quad [A^n = A_{21} = n \times 2]$

Now, $B = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$

$$\Rightarrow B^2 = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow B^2 = \begin{bmatrix} 1+0 & 3+3 \\ 0+0 & 0+1 \end{bmatrix} = \begin{bmatrix} 1 & 6 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow B^3 = \begin{bmatrix} 1 & 6 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 9 \\ 0 & 1 \end{bmatrix}$$

Similarly, $B^6 = \begin{bmatrix} 1 & 18 \\ 0 & 1 \end{bmatrix} \quad [B^n = B_{12} = n \times 3]$

$$\therefore (A^6 + B^6) = \begin{bmatrix} 1 & 0 \\ 12 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 18 \\ 0 & 1 \end{bmatrix}$$

$$A^6 + B^6 = \begin{bmatrix} 2 & 18 \\ 12 & 2 \end{bmatrix}$$

$$\text{Det}(A^6 + B^6) = 4 - 216 = -212$$

Question58

Let $G(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$. If $x + y = 0$ then $G(x)G(y) =$

AP EAPCET 2022 - 5th July Morning Shift

Options:

A. null Matrix

B. skew-symmetric Matrix

C. identity Matrix

D. symmetric Matrix

Answer: C

Solution:

$$\text{Here, } G(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$G(y) = \begin{bmatrix} \cos y & -\sin y & 0 \\ \sin y & \cos y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{So, } x + y = 0 \Rightarrow y = (-x)$$

$$\text{Now, } G(-x) = \begin{bmatrix} \cos(-x) & -\sin(-x) & 0 \\ \sin(-x) & \cos(-x) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} G(x) \cdot G(y) &= \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos x & \sin x & 0 \\ -\sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I \end{aligned}$$

Question59

$$\text{If } A = \begin{bmatrix} 2 & -3 \\ -4 & 1 \end{bmatrix}, \text{ then } (A^T)^2 + (12A)^T =$$

AP EAPCET 2022 - 4th July Evening Shift

Options:

A. $5 \begin{bmatrix} 8 & 12 \\ -9 & 5 \end{bmatrix}$

B. $5 \begin{bmatrix} 8 & -9 \\ -12 & 5 \end{bmatrix}$

C. $\begin{bmatrix} 40 & -45 \\ 60 & 25 \end{bmatrix}$

D. $\begin{bmatrix} 40 & -60 \\ -45 & 25 \end{bmatrix}$

Answer: D

Solution:

Given, $A = \begin{bmatrix} 2 & -3 \\ -4 & 1 \end{bmatrix}$

$$\begin{aligned} (A^T)^2 + (12A)^T &= \begin{bmatrix} 2 & -4 \\ -3 & 1 \end{bmatrix}^2 + \begin{bmatrix} 24 & -36 \\ -48 & 12 \end{bmatrix}^T \\ &= \begin{bmatrix} 16 & -12 \\ -9 & 13 \end{bmatrix} + \begin{bmatrix} 24 & -48 \\ -36 & 12 \end{bmatrix} \\ &= \begin{bmatrix} 16+24 & -12-48 \\ -9-36 & 13+12 \end{bmatrix} = \begin{bmatrix} 40 & -60 \\ -45 & 25 \end{bmatrix} \end{aligned}$$

Question60

If a, b, c are respectively the 5 th, 8 th, 13 th terms of an arithmetic

progression, then $\begin{vmatrix} a & 5 & 1 \\ b & 8 & 1 \\ c & 13 & 1 \end{vmatrix} =$

AP EAPCET 2022 - 4th July Evening Shift

Options:

A. 0

B. 1

C. abc

D. 520

Answer: A

Solution:

Given, a, b and c are respectively 5th, 8th, and 13th terms of arithmetic progression.

$$\therefore a = P + (5 - 1)d$$

$$\Rightarrow a = P + 4d$$

$$b = P + (8 - 1)d$$

$$\Rightarrow b = P + 7d$$

$$c = P + (13 - 1)d$$

$$\Rightarrow c = P + 12d$$

$$\therefore \begin{vmatrix} a & 5 & 1 \\ b & 8 & 1 \\ c & 13 & 1 \end{vmatrix} = \begin{vmatrix} P + 4d & 5 & 1 \\ P + 7d & 8 & 1 \\ P + 12d & 13 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} P & 5 & 1 \\ P & 8 & 1 \\ P & 13 & 1 \end{vmatrix} + \begin{vmatrix} 4d & 5 & 1 \\ 7d & 8 & 1 \\ 12d & 13 & 1 \end{vmatrix}$$

$$= P \begin{vmatrix} 1 & 5 & 1 \\ 1 & 8 & 1 \\ 1 & 13 & 1 \end{vmatrix} + d \begin{vmatrix} 4 + 1 & 5 & 1 \\ 7 + 1 & 8 & 1 \\ 12 + 1 & 13 & 1 \end{vmatrix}$$

$$= P \cdot 0 + d \begin{vmatrix} 5 & 5 & 1 \\ 8 & 8 & 1 \\ 13 & 13 & 1 \end{vmatrix} = P \cdot 0 + d \cdot 0 = 0$$

Question61

If $A = \begin{bmatrix} 1 & 0 & 0 \\ a & -1 & 0 \\ b & c & 1 \end{bmatrix}$ is such that $A^2 = I$, then

AP EAPCET 2022 - 4th July Evening Shift

Options:

A. $b = \frac{ac}{2}$

B. $b = -\frac{ac}{2}$

C. $b = \frac{a+c}{2}$

D. $b = \sqrt{ac}$

Answer: B

Solution:

Given that,

$$A = \begin{bmatrix} 1 & 0 & 0 \\ a & -1 & 0 \\ b & c & 1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ a & -1 & 0 \\ b & c & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ a & -1 & 0 \\ b & c & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ b+ca+b & 0 & 1 \end{bmatrix}$$

Since, $A^2 = I$ (given)

$$\Rightarrow b + ca + b = 0$$

$$\Rightarrow b = \frac{-ca}{2}$$

Question62

Let $A = \begin{bmatrix} -2 & x & 1 \\ x & 1 & 1 \\ 2 & 3 & -1 \end{bmatrix}$. If the roots of the equation $\det A = 0$ are l, m then $l^3 - m^3 =$

AP EAPCET 2022 - 4th July Evening Shift

Options:

A. 35

B. -35

C. 19

D. -19

Answer: C

Solution:

Given,

$$A = \begin{bmatrix} -2 & x & 1 \\ x & 1 & 1 \\ 2 & 3 & -1 \end{bmatrix}$$

$$\text{Given, } \det A = \begin{vmatrix} -2 & x & 1 \\ x & 1 & 1 \\ 2 & 3 & -1 \end{vmatrix} = 0$$

$$\begin{aligned}
&\Rightarrow -2(-1-3) + x(2+x) + 1(3x-2) = 0 \\
&\Rightarrow 8 + 2x + x^2 + 3x - 2 = 0 \\
&\Rightarrow x^2 + 5x + 6 = 0 \Rightarrow (x+2)(x+3) = 0 \\
&\Rightarrow x = -2 \text{ and } -3 \\
&\therefore l = -2 \text{ and } m = -3 \\
&\quad l^3 - m^3 = (-2)^3 - (-3)^3 = -8 + 27 = 19
\end{aligned}$$

Question 63

For $i = 1, 2, 3$ and $j = 1, 2, 3$ If

$$a_i^2 + b_i^2 + c_i^2 = 1, a_i a_j + b_i b_j + c_i c_j = 0, \forall i \neq j \text{ and}$$

$$A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}, \text{ then } \det(AA^T) =$$

AP EAPCET 2022 - 4th July Morning Shift

Options:

A. 0

B. 1

C. -1

D. 3

Answer: B

Solution:

$$\text{Given, } a_i^2 + b_i^2 + c_i^2 = 1$$

$$\Rightarrow a_1^2 + b_1^2 + c_1^2 = 1$$

$$a_2^2 + b_2^2 + c_2^2 = 1$$

$$a_3^2 + b_3^2 + c_3^2 = 1$$

$$\text{Given, } a_i a_j + b_i b_j + c_i c_j = 0 \quad [i, j = 1, 2, 3, i \neq j]$$

$$\text{Let } \mathbf{p} = a_1 \mathbf{i} + b_1 \mathbf{j} + c_1 \mathbf{k}$$

$$\mathbf{q} = a_2 \mathbf{i} + b_2 \mathbf{j} + c_2 \mathbf{k}$$

$$\mathbf{r} = a_3 \mathbf{i} + b_3 \mathbf{j} + c_3 \mathbf{k}$$

$$|\mathbf{p}| = |\mathbf{q}| = |\mathbf{r}| = 1$$

$$\mathbf{p} \cdot \mathbf{q} = (a_1 \mathbf{i} + b_1 \mathbf{j} + c_1 \mathbf{k}) \cdot (a_2 \mathbf{i} + b_2 \mathbf{j} + c_3 \mathbf{k})$$

$$= a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$$

$$\Rightarrow \mathbf{p} \perp \mathbf{q}$$

Similarly, $\mathbf{q} \perp \mathbf{r}, \mathbf{r} \perp \mathbf{p}$

$$\text{Now, } A = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = [\mathbf{p} \quad \mathbf{q} \quad \mathbf{r}] = 1$$
$$\Rightarrow \det(AA^T) = 1$$

Question 64

If $A = \frac{1}{7} \begin{bmatrix} 3 & -2 & 6 \\ -6 & -3 & 2 \\ -2 & 6 & 3 \end{bmatrix}$, then

AP EAPCET 2022 - 4th July Morning Shift

Options:

A. $A^{-1} = A$

B. $A^{-1} = A^T$

C. A^{-1} does not exist

D. $A^{-1} = -A$

Answer: B

Solution:

$$A = \frac{1}{7} \begin{bmatrix} 3 & -2 & 6 \\ -6 & -3 & 2 \\ -2 & 6 & 3 \end{bmatrix}$$
$$|A| = \frac{1}{7^3} \begin{vmatrix} 3 & -2 & 6 \\ -6 & -3 & 2 \\ -2 & 6 & 3 \end{vmatrix} \quad [\because \det(kA) = k^n \det A]$$
$$= \frac{1}{7^3} [3(-9 - 12) + 2(-18 + 4) + 6(-36 - 6)]$$
$$= \frac{1}{7^3} [-63 - 28 - 252]$$
$$= \frac{-343}{7^3} = -1 \neq 0$$

$$\text{Adj}(A) = \frac{1}{7^2} \begin{bmatrix} -21 & 14 & -42 \\ 42 & 21 & -14 \\ 14 & -42 & -21 \end{bmatrix}$$

$$[\because \operatorname{adj}(kA) = k^{n-1} \operatorname{adj} A]$$

$$= \frac{1}{7^2} \begin{bmatrix} -21 & 42 & 14 \\ 14 & 21 & -42 \\ -42 & -14 & -21 \end{bmatrix}$$

$$\text{Now, } A^{-1} = \frac{\operatorname{adj}(A)}{|A|} = \frac{-1}{7^2} \begin{bmatrix} -21 & 42 & 14 \\ 14 & 21 & -42 \\ -42 & -14 & -21 \end{bmatrix}$$

$$= \frac{1}{7} \begin{bmatrix} 3 & -6 & -2 \\ -2 & -3 & 6 \\ 6 & 2 & 3 \end{bmatrix}$$

$$A^{-1} = A^T$$

Question65

If $A = \begin{bmatrix} \alpha^2 & 5 \\ 5 & -\alpha \end{bmatrix}$ and $\det(A^{10}) = 1024$, then $\alpha =$

AP EAPCET 2022 - 4th July Morning Shift

Options:

A. -2

B. -1

C. -3

D. 0

Answer: C

Solution:

Step 1: Given matrix A

The matrix A is given as:

$$A = \begin{bmatrix} \alpha^2 & 5 \\ 5 & -\alpha \end{bmatrix}$$

Step 2: Find the determinant of A

The determinant of a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is calculated by:

$$\det(A) = ad - bc$$

For matrix A , we have:

$$a = \alpha^2, \quad b = 5, \quad c = 5, \quad d = -\alpha$$

So,

$$\det(A) = (\alpha^2)(-\alpha) - (5)(5) = -\alpha^3 - 25$$

Thus, the determinant of A is:

$$\det(A) = -\alpha^3 - 25$$

Step 3: Use the condition $\det(A^{10}) = 1024$

Now, we are given that $\det(A^{10}) = 1024$. The property of determinants tells us that:

$$\det(A^n) = (\det(A))^n$$

Thus:

$$\det(A^{10}) = (\det(A))^{10}$$

We are told that:

$$(\det(A))^{10} = 1024$$

Since $1024 = 2^{10}$, we have:

$$(\det(A))^{10} = 2^{10}$$

Taking the 10th root of both sides:

$$\det(A) = 2$$

Step 4: Solve for α

From earlier, we found that:

$$\det(A) = -\alpha^3 - 25$$

We know that $\det(A) = 2$, so:

$$-\alpha^3 - 25 = 2$$

Solving for α^3 :

$$-\alpha^3 = 27$$

$$\alpha^3 = -27$$

Taking the cube root of both sides:

$$\alpha = -3$$

Final Answer:

$$\boxed{-3}$$

Question66

Let $A = \begin{bmatrix} 5 & \sin^2 \theta & \cos^2 \theta \\ -\sin^2 \theta & -5 & 1 \\ \cos^2 \theta & 1 & 5 \end{bmatrix}$. Then, maximum value of $\det(A)$ is

AP EAPCET 2022 - 4th July Morning Shift

Options:

A. -125

B. 200

C. $-\frac{255}{2}$

D. 145

Answer: A

Solution:

$$A = \begin{bmatrix} 5 & \sin^2 \theta & \cos^2 \theta \\ -\sin^2 \theta & -5 & 1 \\ \cos^2 \theta & 1 & 5 \end{bmatrix}$$

$$\begin{aligned} |A| &= 5(-25 - 1) - \sin^2 \theta (-5 \sin^2 \theta - \cos^2 \theta) \\ &+ \cos^2 \theta (-\sin^2 \theta + 5 \cos^2 \theta) \\ &= -130 - \sin^2 \theta [-5(1 - \cos^2 \theta) - \cos^2 \theta] \\ &+ \cos^2 \theta [-\sin^2 \theta + 5(1 - \sin^2 \theta)] \\ &= -130 - \sin^2 \theta (-5 + 4 \cos^2 \theta) + \cos^2 \theta (-6 \sin^2 \theta + 5) \\ &= -130 + 5 \sin^2 \theta - 4 \sin^2 \theta \cos^2 \theta \\ &\quad - 6 \sin^2 \theta \cos^2 \theta + 5 \cos^2 \theta \\ &= -130 + 5 - 10 \sin^2 \theta \cos^2 \theta \\ |A| &= -125 - 10 \sin^2 \theta \cos^2 \theta \\ \therefore |A|_{\max} &= -125 \end{aligned}$$

Question 67

If $\frac{x^4+24x^2+28}{(x^2+1)^3} = \frac{Ax+B}{x^2+1} + \frac{Cx+D}{(x^2+1)^2} + \frac{Ex+F}{(x^2+1)^3}$, then the value of $A + B + C + D + E + F =$

AP EAPCET 2022 - 4th July Morning Shift

Options:

A. 21

B. 22

C. 28

D. 29

Answer: C

Solution:

$$\frac{x^4+24x^2+28}{(x^2+1)^3} = \frac{Ax+B}{x^2+1} + \frac{Cx+D}{(x^2+1)^2} + \frac{Ex+F}{(x^2+1)^3}$$

$$\text{Let } x^2 + 1 = k$$

$$\Rightarrow x^2 = k - 1$$

Substituting $x^2 = k - 1$ in LHS, we get

$$\begin{aligned}\frac{(k-1)^2 + 24(k-1) + 28}{k^3} &= \frac{k^2 + 22k + 5}{k^3} \\&= \frac{1}{k} + \frac{22}{k^2} + \frac{5}{k^3} \\&= \frac{1}{(x^2+1)} + \frac{22}{(x^2+1)^2} + \frac{5}{(x^2+1)^3}\end{aligned}$$

On comparing with RHS, we get

$$Ax + B = 1, Cx + D = 22 \text{ and } Ex + F = 5$$

$$\text{At } x = 1,$$

$$A + B = 1, C + D = 22 \text{ and } E + F = 5$$

$$\therefore A + B + C + D + E + F = 1 + 22 + 5 = 28$$

Question68

If $k \in R$ and $\det A = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = k$, then

$\det B = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 + 2a_1 & b_2 + 2b_1 & c_2 + 2c_1 \\ a_3 & b_3 & c_3 \end{vmatrix}$ is equal to

AP EAPCET 2021 - 20th August Evening Shift

Options:

A. 0

B. 2k

C. k

D. k^2

Answer: C

Solution:

$$\det A = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = k \text{ (given)}$$

$$\begin{aligned}
 \text{Then, } \det B &= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 + 2a_1 & b_2 + 2b_1 & c_2 + 2c_1 \\ a_3 & b_3 & c_3 \end{vmatrix} \\
 &= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} a_1 & b_1 & c_1 \\ 2a_1 & 2b_1 & 2c_1 \\ a_3 & b_3 & c_3 \end{vmatrix} \\
 &= k + 0 \\
 \Rightarrow \det B &= k
 \end{aligned}$$

Question69

If $A = \begin{bmatrix} \sqrt{2020} & \sqrt{2021} & \sqrt{2021} & \sqrt{2023} \\ \sqrt{4040} & \sqrt{4042} & \sqrt{4044} & \sqrt{4046} \\ \sqrt{6060} & \sqrt{6063} & \sqrt{6066} & \sqrt{6069} \\ \sqrt{8080} & \sqrt{8084} & \sqrt{8088} & \sqrt{8092} \end{bmatrix}$, then the rank of A is

AP EAPCET 2021 - 20th August Evening Shift

Options:

- A. 1
- B. 2
- C. 3
- D. 4

Answer: A

Solution:

$$\begin{aligned}
 A &= \begin{bmatrix} \sqrt{2020} & \sqrt{2022} & \sqrt{2022} & \sqrt{2023} \\ \sqrt{4040} & \sqrt{4042} & \sqrt{4044} & \sqrt{4046} \\ \sqrt{6060} & \sqrt{6063} & \sqrt{6066} & \sqrt{6069} \\ \sqrt{8080} & \sqrt{8084} & \sqrt{8088} & \sqrt{8092} \end{bmatrix} \\
 &= \begin{bmatrix} \sqrt{2020} & \sqrt{2021} & \sqrt{2022} & \sqrt{2023} \\ \sqrt{2} \cdot \sqrt{2020} & \sqrt{2}\sqrt{2021} & \sqrt{4044} & \sqrt{4046} \\ \sqrt{3}\sqrt{2020} & \sqrt{3}\sqrt{2021} & \sqrt{6066} & \sqrt{6069} \\ \sqrt{4}\sqrt{2020} & \sqrt{4}\sqrt{2021} & \sqrt{8088} & \sqrt{8092} \end{bmatrix} \\
 R_2 &\rightarrow R_2 - \sqrt{2}R_1, R_3 \rightarrow R_3 - \sqrt{3}R_1, R_4 \rightarrow R_4 - \sqrt{4}R_1 \\
 A &\sim \begin{bmatrix} \sqrt{2020} & \sqrt{2021} & \sqrt{2022} & \sqrt{2023} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}
 \end{aligned}$$

$\therefore$ A has one non-zero row

$\therefore$ Rank (A) = 1

Option (a) is correct.

Question70

If $\begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0$ and x, y and z are all distinct, then xyz is equal to

AP EAPCET 2021 - 20th August Evening Shift

Options:

- A. -1
- B. 1
- C. 0
- D. 3

Answer: A

Solution:

The given equation is:

$$\begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0$$

We can rewrite the third column as $(1) + (\text{the cube term})$, so this is the same as:

$$\begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ y & y^2 & y^3 \\ z & z^2 & z^3 \end{vmatrix} = 0$$

Let's call the first determinant as A and the second as B .

Step 1: Find the value of $|A|$

$$\text{Let: } A = \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix}$$

$$\text{Subtract first row from second and third, } R_2 \rightarrow R_2 - R_1, \quad R_3 \rightarrow R_3 - R_1 \text{ so, } \begin{vmatrix} x & x^2 & 1 \\ y-x & y^2-x^2 & 0 \\ z-x & z^2-x^2 & 0 \end{vmatrix}$$

Now, since last column for R_2 and R_3 is 0, expand along third column:

$$\begin{aligned} |A| &= (y-x)(z^2-x^2) - (z-x)(y^2-x^2) \\ &= (y-x)(z-x)(z+x) - (z-x)(y-x)(y+x) \\ &= (y-x)(z-x)(z+x-y-x) \\ &= (y-x)(z-x)(z-y) \end{aligned}$$

Step 2: Find the value of $|B|$

$$B = \begin{vmatrix} x & x^2 & x^3 \\ y & y^2 & y^3 \\ z & z^2 & z^3 \end{vmatrix}$$

$$\text{Factor } x, y, z \text{ from each row: } = xyz \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$$

Now, do the same row operation as before: $R_2 \rightarrow R_2 - R_1, \quad R_3 \rightarrow R_3 - R_1$

$$\text{So determinant becomes: } xyz \begin{vmatrix} 1 & x & x^2 \\ 0 & y-x & y^2-x^2 \\ 0 & z-x & z^2-x^2 \end{vmatrix}$$

Expand this along the first column:

$$\begin{aligned} |B| &= xyz \{ (y-x)(z^2-x^2) - (z-x)(y^2-x^2) \} \\ &= xyz(y-x)(z-x)(z-y) \end{aligned}$$

Step 3: Use the original equation

We have: $|A| + |B| = 0$ So, $(y-x)(z-x)(z-y) + xyz(y-x)(z-x)(z-y) = 0$

Factor out $(y-x)(z-x)(z-y)$: $(y-x)(z-x)(z-y)(1+xyz) = 0$

Because x, y, z are all different, $(y - x)(z - x)(z - y) \neq 0$. So, $1 + xyz = 0$

Therefore, $xyz = -1$

Question 71

Let A be a $n \times n$ matrix such that A is upper-triangular. Then, $\text{adj}(A)$ is equal to

AP EAPCET 2021 - 20th August Evening Shift

Options:

- A. lower triangular matrix
- B. upper triangular matrix
- C. diagonal matrix
- D. scalar matrix

Answer: B

Solution:

Let $n = 2$ (say) and $A = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$, Then $\text{adj } A = \begin{bmatrix} c & -b \\ 0 & a \end{bmatrix}$ which is a upper triangular matrix. Then, for any $n \times n$ matrix A , $\text{adj } A$ is also upper triangular matrix.

Question 72

If $f(x) = \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix}$, then the ratio $f''(x) : f'(x)$ is equal to

AP EAPCET 2021 - 20th August Evening Shift

Options:

- A. $2 : x$

B. $x^2 : x$

C. $3x : 2$

D. $6 : x$

Answer: A

Solution:

$$f(x) = \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix}$$

Expand along column 1

$$\begin{aligned} f(x) &= x(12x^2 - 6x^2) - (6x^3 - 2x^3) \\ &= x(6x^2) - 4x^3 = 6x^3 - 4x^3 = 2x^3 \\ \therefore f(x) &= 2x^3 \Rightarrow f'(x) = 6x^2 \\ f''(x) &= 12x \end{aligned}$$

Then, $f''(x) : f'(x) = 12x : 6x^2 = 2 : x$

Question73

The trace of the matrix $A = \begin{bmatrix} 1 & -5 & 7 \\ 0 & 7 & 9 \\ 11 & 8 & 9 \end{bmatrix}$ is

AP EAPCET 2021 - 20th August Morning Shift

Options:

A. 17

B. 25

C. 3

D. 12

Answer: A

Solution:

Trace of A = Sum of diagonal elements of A

$$\text{Given, } A = \begin{bmatrix} 1 & -5 & 7 \\ 0 & 7 & 9 \\ 11 & 8 & 9 \end{bmatrix}$$

$$= 1 + 7 + 9 = 17$$

Question 74

If A, B and C are the angles of a triangle, then the system of equations $-x + y \cos C + z \cos B = 0$, $x \cos C - y + z \cos A = 0$ and $x \cos B + y \cos A - z = 0$

AP EAPCET 2021 - 20th August Morning Shift

Options:

- A. Only zero solution
- B. A non-zero solution for all ΔABC
- C. Only zero solution but for certain values of A, B and C
- D. A non-zero solution if ΔABC is an equilateral triangle and not for all triangles.

Answer: B

Solution:

$$-x + y \cos C + z \cos B = 0$$

$$x \cos C - y + z \cos A = 0$$

$$x \cos B + y \cos A - z = 0$$

$$\Delta = \begin{vmatrix} -1 & \cos C & \cos B \\ \cos C & -1 & \cos A \\ \cos B & \cos A & -1 \end{vmatrix}$$

$$= -1(1 - \cos^2 A) + \cos C(\cos A \cos B + \cos C) \\ + \cos B(\cos A \cos C + \cos B)$$

$$= -1 + \cos^2 A + 2 \cos A \cos B \cos C \\ + \cos^2 C + \cos^2 B$$

$$= -\sin^2 A + \cos^2 B + \cos^2 C + 2 \cos A \cos B \cos C$$

Clearly, $\Delta > 0$.

$\Rightarrow$ System of equations has a non-zero solution.

Question75

If $\begin{bmatrix} 1 & -\tan \theta \\ \tan \theta & 1 \end{bmatrix} \begin{bmatrix} 1 & \tan \theta \\ -\tan \theta & 1 \end{bmatrix}^{-1} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, then

AP EAPCET 2021 - 20th August Morning Shift

Options:

- A. $a = 1, b = 1$
- B. $a = \sin 2\theta$ and $b = \cos 2\theta$
- C. $a = \cos 2\theta$ and $b = \sin 2\theta$
- D. $a = 0$ and $b = 0$

Answer: C

Solution:

$$\begin{aligned} \text{Let } A &= \begin{bmatrix} 1 & \tan \theta \\ -\tan \theta & 1 \end{bmatrix} \\ A^{-1} &= \frac{\text{adj}(A)}{|A|} \\ &= \left(\frac{1}{\sec^2 \theta} \right) \begin{bmatrix} 1 & -\tan \theta \\ \tan \theta & 1 \end{bmatrix} \\ \text{So, } \frac{1}{\sec^2 \theta} \begin{bmatrix} 1 & -\tan \theta \\ \tan \theta & 1 \end{bmatrix} \begin{bmatrix} 1 & -\tan \theta \\ \tan \theta & 1 \end{bmatrix} \\ &= \frac{1}{\sec^2 \theta} \begin{bmatrix} 1 - \tan^2 \theta & -2 \tan \theta \\ 2 \tan \theta & -\tan^2 \theta + 1 \end{bmatrix} \\ \begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix} &= \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \\ \Rightarrow a &= \cos 2\theta, b = \sin 2\theta \end{aligned}$$

Question76

What is the value of $\begin{vmatrix} a & b & c \\ a-b & b-c & c-a \\ b+c & c+a & a+b \end{vmatrix}$?

AP EAPCET 2021 - 20th August Morning Shift

Options:

A. $a^3 + b^3 + c^3 + 3abc$

B. $a^3 + b^3 + c^3 - 3abc$

C. $a^3 + b^3 + c^3 - 6abc$

D. $a^3 + b^3 + c^3 + 6abc$

Answer: B

Solution:

$$\begin{aligned} & \begin{bmatrix} a & b & c \\ a-b & b-c & c-a \\ b+c & c+a & a+b \end{bmatrix} \quad [R_1 \rightarrow R_1 + R_3] \\ &= (a+b+c) \begin{bmatrix} 1 & 1 & 1 \\ a-b & b-c & c-a \\ b+c & c+a & a+b \end{bmatrix} \\ &= (a+b+c) \begin{bmatrix} 1 & 1 & 1 \\ a+c & b+a & b+c \\ b+c & c+a & a+b \end{bmatrix} \quad [R_2 \rightarrow R_2 + R_3] \end{aligned}$$

$$\begin{aligned} & C_2 \rightarrow C_2 - C_1 \text{ and } C_3 \rightarrow C_3 - C_1 \\ &= (a+b+c) \begin{bmatrix} 1 & 0 & 0 \\ a+c & b-c & b-a \\ b+c & a-b & a-c \end{bmatrix} \\ &= (a+b+c)[(b-c)(a-c) - (b-a)(a-b)] \\ &= (a+b+c)(ab - bc - ca + c^2 - ba + b^2 + a^2 - ab) \\ &= (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca) \\ &= a^3 + b^3 + c^3 - 3abc \end{aligned}$$

Question 77

The value of $\begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$ is

AP EAPCET 2021 - 19th August Evening Shift

Options:

A. abc

B. $(a+b)(b+c)(c+a)$

C. $4abc$

D. $(a-b)(b-c)(c-a)$

Answer: C

Solution:

$$\begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix} \\ = (b+c)[(c+a)(a+b) - bc] - a(ab + b^2 - bc) \\ = (b+c)(ac + bc + a^2 + ab - bc) + a(bc - c^2 - ac) \\ - a^2b - ab^2 + abc + abc - ac^2 - a^2c \\ = abc + a^2b + ab^2 + ac^2 + a^2c + abc - a^2b - ab^2 + abc + abc - ac^2 - a^2c \\ = 4abc$$

Option (c) is correct.

Question 78

Let A, B, C, D be square real matrices such that $C^T = DAB$, $D^T = ABC$ and $S = ABCD$, then S^2 is equal to

AP EAPCET 2021 - 19th August Evening Shift

Options:

A. S

B. BCD

C. S^T

D. $(S^T)^2 = (S^2)^T$

Answer: D

Solution:

$$D^T = ABC$$

$$\text{and } S = ABCD$$

$$S^2 = (ABCD)(ABCD) = (D^T \cdot D)(D^T D)$$

$$= (D^T D)^2 \dots (i)$$

$$\therefore S = ABCD = D^T D$$

$$S^T = (D^T D)^T = D^T D$$

$\therefore$ From Eq. (i), we get

$$S^2 = (S^T)^2$$

$$\therefore S^2 = (S^T)^2 = (S^2)^T$$

Question79

$$A = \begin{bmatrix} a^2 & 15 & 31 \\ 12 & b^2 & 41 \\ 35 & 61 & c^2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2a & 3 & 5 \\ 2 & 2b & 8 \\ 1 & 4 & 2c - 3 \end{bmatrix} \text{ are two matrices}$$

such that the sum of the principal diagonal elements of both A and B are equal, then the product of the principal diagonal elements of B is

AP EAPCET 2021 - 19th August Evening Shift

Options:

A. 4

B. 0

C. -4

D. -12

Answer: C

Solution:

$$A = \begin{bmatrix} a^2 & 15 & 31 \\ 12 & b^2 & 41 \\ 35 & 61 & c^2 \end{bmatrix},$$

$$B = \begin{bmatrix} 2a & 3 & 5 \\ 2 & 2b & 8 \\ 1 & 4 & 2c - 3 \end{bmatrix}$$

Given, trace of A = trace of B

$$\begin{aligned} \Rightarrow a^2 + b^2 + c^2 &= 2a + 2b + 2c - 3 \\ \Rightarrow (a^2 - 2a + 1) + (b^2 - 2b + 1) + (c^2 - 2c + 1) &= 0 \\ \Rightarrow (a - 1)^2 + (b - 1)^2 + (c - 1)^2 &= 0 \\ \Rightarrow a = 1, b = 1, c = 1 \end{aligned}$$

Then product of diagonal elements of B

$$= (2a)(2b)(2c - 3) = (2)(2)(2 - 3) = -4$$

Question80

Let a, b and c be such that $b + c \neq 0$ and

$$\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ (-1)^{n+2}a & (-1)^{n-1}b & (-1)^nc \end{vmatrix} = 0,$$

then the value of n is

AP EAPCET 2021 - 19th August Evening Shift

Options:

- A. zero
- B. any even integer
- C. any odd integer
- D. any integer

Answer: C

Solution:

$$\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix}$$

$$+ \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ (-1)^{n+2}a & (-1)^{n-1}b & (-1)^n c \end{vmatrix} = 0$$

$$\because |A| = |A^T|$$

$$\Rightarrow \begin{vmatrix} a & -b & c \\ a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \end{vmatrix} + (-1)^n \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ a & -b & c \end{vmatrix} = 0$$

$R_2 \leftrightarrow R_3$ and $R_3 \leftrightarrow R_1$ in 2nd determinant

$$\Rightarrow \begin{vmatrix} a & -b & c \\ a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \end{vmatrix} + (-1)^n \begin{vmatrix} a & -b & c \\ a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \end{vmatrix} = 0$$

Both determinants are equal and their sum is zero when n is an odd integer.

Question 81

The equation whose roots are the values of the equation

$$\begin{vmatrix} 1 & -3 & 1 \\ 1 & 6 & 4 \\ 1 & 3x & x^2 \end{vmatrix} = 0 \text{ is}$$

AP EAPCET 2021 - 19th August Morning Shift

Options:

A. $x^2 + x + 2 = 0$

B. $x^2 + x - 2 = 0$

C. $x^2 + 2x + 2 = 0$

D. $x^2 - x - 2 = 0$

Answer: D

Solution:

$$\begin{vmatrix} 1 & -3 & 1 \\ 1 & 6 & 4 \\ 1 & 3x & x^2 \end{vmatrix} = 0$$
$$\Rightarrow 1(6x^2 - 12x) + 3(x^2 - 4) + 1(3x - 6) = 0$$
$$\Rightarrow 6x^2 - 12x + 3x^2 - 12 + 3x - 6 = 0$$
$$\Rightarrow 9x^2 - 9x - 18 = 0$$
$$\Rightarrow x^2 - x - 2 = 0$$

Question82

Let **a** and **b** be non-zero real numbers such that $ab = 5/2$ and given $A = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ and $AA^T = 20I$ (I is unit matrix), then the equation whose roots are **a** and **b** is

AP EAPCET 2021 - 19th August Morning Shift

Options:

A. $x^2 \mp 10x + 5 = 0$

B. $2x^2 \pm 10x + 5 = 0$

C. $x^2 - 5x + \frac{5}{2} = 0$

D. $x^2 - 25x + \frac{5}{2} = 0$

Answer: B

Solution:

$$A = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

$$\text{Then, } A^T = \begin{bmatrix} a & b \\ -b & a \end{bmatrix}$$

$$\therefore AA^T = 20I$$

$$\Rightarrow \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \begin{bmatrix} a & b \\ -b & a \end{bmatrix} = 20 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a^2 + b^2 & 0 \\ 0 & a^2 + b^2 \end{bmatrix} = \begin{bmatrix} 20 & 0 \\ 0 & 20 \end{bmatrix}$$

$$\therefore a^2 + b^2 = 20$$

$$\Rightarrow (a + b)^2 - 2ab = 20$$

$$\Rightarrow (a + b)^2 = 20 + 2 \times \frac{5}{2} \quad \left[\because ab = \frac{5}{2} \right]$$

$$\text{or } (a + b)^2 = 25$$

$$\text{or } a + b = \pm 5$$

Equation whose roots are a and b is given by $x^2 + (a + b)x + ab = 0$

$$\text{or } x^2 \pm 5x + \frac{5}{2} = 0 \text{ or } 2x^2 \pm 10x + 5 = 0$$

Question83

If $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$, $10B = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & -2 & 3 \end{bmatrix}$ and $B = A^{-1}$, then the value of α is

AP EAPCET 2021 - 19th August Morning Shift

Options:

A. 2

B. 0

C. 5

D. 4

Answer: C

Solution:

$$(c) A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\therefore |A| = 1(1 + 3) + 1(2 + 3) + 1(2 - 1) = 10 \quad \dots (i)$$

$$\text{and } 10B = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & -2 & 3 \end{bmatrix}$$

$$\therefore |10B| = 4(2\alpha) - 2(-15 - \alpha) + 2(10)$$

$$\Rightarrow 10^3 |B| = 10\alpha + 50 \quad [\because |kA| = k^n |A|] \dots (ii)$$

$$\therefore B = A^{-1}$$

$$\Rightarrow |B| = |A^{-1}| = |A|^{-1}$$

$$\Rightarrow |B| = \frac{1}{|A|} = \frac{1}{10} \quad [\text{from Eq. (i)}]$$

[from Eq. (i)] From Eq. (ii), we get

$$1000 \cdot \frac{1}{10} = 10\alpha + 50$$

$$\text{or } \alpha = \frac{50}{10} = 5$$

Question 84

The rank of the matrix $\begin{bmatrix} 4 & 2 & (1-x) \\ 5 & k & 1 \\ 6 & 3 & (1+x) \end{bmatrix}$ is 1, then,

AP EAPCET 2021 - 19th August Morning Shift

Options:

A. $k = \frac{5}{2}, x = \frac{1}{5}$

B. $k = \frac{5}{2}, x \neq \frac{1}{5}$

C. $k = \frac{1}{5}, x = \frac{5}{2}$

D. $k \neq \frac{5}{2}, x = \frac{1}{5}$

Answer: A

Solution:

$$\text{Let } A = \begin{bmatrix} 4 & 2 & 1-x \\ 5 & k & 1 \\ 6 & 3 & 1+x \end{bmatrix}$$

Given that, rank of matrix A is 1 .

$\Rightarrow$ All the minors of order 2 are zero.

$$\therefore M_{33} = 0$$

$$\Rightarrow \begin{vmatrix} 4 & 2 \\ 5 & k \end{vmatrix} = 0$$

$$\text{or } 4k - 10 = 0 \text{ or } k = \frac{5}{2}$$

$$\text{and } M_{11} = \begin{vmatrix} k & 1 \\ 3 & 1+x \end{vmatrix} = 0$$

$$\text{or } \begin{vmatrix} \frac{5}{2} & 1 \\ 3 & 1+x \end{vmatrix} = 0$$

$$\text{or } \frac{5}{2} + \frac{5}{2}x - 3 = 0$$

$$\text{or } \frac{5}{2}x = \frac{1}{2}$$

$$\text{or } x = \frac{1}{5}$$

Question85

If $a_1, a_2, \dots, a_9$ are in GP, then $\begin{vmatrix} \log a_1 & \log a_2 & \log a_3 \\ \log a_4 & \log a_5 & \log a_6 \\ \log a_7 & \log a_8 & \log a_9 \end{vmatrix}$ is equal to

AP EAPCET 2021 - 19th August Morning Shift

Options:

A. $\log(a_1, a_2, \dots, a_n)$

B. 1

C. $(\log a_9)^9$

D. 0

Answer: D

Solution:

$a_1, a_2, \dots, a_9$, are in GP.

Then, $\frac{a_2}{a_1} = \frac{a_3}{a_2} = \dots = \frac{a_9}{a_8} = r$ [common ratio]

$$\text{Now, } \begin{vmatrix} \log a_1 & \log a_2 & \log a_3 \\ \log a_4 & \log a_5 & \log a_6 \\ \log a_7 & \log a_8 & \log a_9 \end{vmatrix}$$

$$C_3 \rightarrow C_3 - C_2, C_2 \rightarrow C_2 - C_1$$

$$\begin{vmatrix} \log a_1 & \log \frac{a_2}{a_1} & \log \frac{a_3}{a_2} \\ \log a_4 & \log \frac{a_5}{a_4} & \log \frac{a_6}{a_5} \\ \log a_7 & \log \frac{a_8}{a_7} & \log \frac{a_9}{a_8} \end{vmatrix}$$

$$\text{or } \begin{vmatrix} \log a_1 & \log r & \log r \\ \log a_4 & \log r & \log r \\ \log a_7 & \log r & \log r \end{vmatrix} = 0$$

[$\because C_2$ and C_3 are identical]

Question86

If $\mathbf{a} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} + 3\hat{\mathbf{k}}$, $\mathbf{b} = \hat{\mathbf{i}} + 3\hat{\mathbf{j}} - \hat{\mathbf{k}}$ and $\mathbf{c} = 3\hat{\mathbf{i}} - \hat{\mathbf{j}} - 2\hat{\mathbf{k}}$, then the value of $\begin{vmatrix} \mathbf{a} \cdot \mathbf{a} & \mathbf{a} \cdot \mathbf{b} & \mathbf{a} \cdot \mathbf{c} \\ \mathbf{b} \cdot \mathbf{a} & \mathbf{b} \cdot \mathbf{b} & \mathbf{b} \cdot \mathbf{c} \\ \mathbf{c} \cdot \mathbf{a} & \mathbf{c} \cdot \mathbf{b} & \mathbf{c} \cdot \mathbf{c} \end{vmatrix}$ is equal to

AP EAPCET 2021 - 19th August Morning Shift

Options:

A. 2020

B. 2025

C. 2030

D. 1849

Answer: B

Solution:

$$\mathbf{a} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} + 3\hat{\mathbf{k}}, \mathbf{b} = \hat{\mathbf{i}} + 3\hat{\mathbf{j}} - \hat{\mathbf{k}}, \mathbf{c} = 3\hat{\mathbf{i}} - \hat{\mathbf{j}} - 2\hat{\mathbf{k}}$$

$$\mathbf{a} \cdot \mathbf{a} = 4 + 1 + 9 = 14$$

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a} = 2 + 3 - 3 = 2$$

$$\mathbf{a} \cdot \mathbf{c} = \mathbf{c} \cdot \mathbf{a} = 6 - 1 - 6 = -1$$

$$\mathbf{b} \cdot \mathbf{b} = 1 + 9 + 1 = 11$$

$$\mathbf{b} \cdot \mathbf{c} = \mathbf{c} \cdot \mathbf{b} = 3 - 3 + 2 = 2$$

$$\mathbf{c} \cdot \mathbf{c} = 9 + 1 + 4 = 14$$

$$\text{Now, } \begin{vmatrix} \mathbf{a} \cdot \mathbf{a} & \mathbf{a} \cdot \mathbf{b} & \mathbf{a} \cdot \mathbf{c} \\ \mathbf{b} \cdot \mathbf{a} & \mathbf{b} \cdot \mathbf{b} & \mathbf{b} \cdot \mathbf{c} \\ \mathbf{c} \cdot \mathbf{a} & \mathbf{c} \cdot \mathbf{b} & \mathbf{c} \cdot \mathbf{c} \end{vmatrix} = \begin{vmatrix} 14 & 2 & -1 \\ 2 & 11 & 2 \\ -1 & 2 & 14 \end{vmatrix}$$

$$= 14(154 - 4) - 2(28 + 2) - 1(4 + 11)$$

$$= 2100 - 60 - 15$$

$$= 2025$$
